Quantifying Decoupling in Global Production and Trade *

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Abstract

We analyze the welfare implications of fragmentation in global trade and multinational production (MP) across countries, across U.S. states, and over time. Combining a semi-endogenous growth model that incorporates subnational regions and technology diffusion through both trade and MP channels with bilateral flow data from 2000 to 2019, we recover changes in bilateral trade and MP costs. Observed cost changes during 2016 to 2019 generate sizable welfare effects with large heterogeneity across countries and U.S. states, with MP cost changes accounting for the dominant share of welfare variation across countries. Endogenous knowledge accumulation amplifies these gains substantially relative to a purely static analysis. We classify countries into China-integrating, US-integrating, or non-integrating groups based on changes in trade and MP costs, finding that countries often maintain inconsistent alignments across the two dimensions and that MP integration patterns are more persistent than trade. Counterfactual exercises show that aligning trade and MP integration with the same group would predominantly reduce welfare, as short-run efficiency gains are offset by impaired technology diffusion in the long run.

Keywords: Economic fragmentation, Multinational production, Technology diffusion

JEL Classification: F14, F23, F43, O33

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1 Introduction

Global economic integration is undergoing a pronounced realignment. Countries are increasingly sorting into groups centered on major powers, and a growing body of research has documented the trade dimensions of this fragmentation (Goldberg and Reed, 2023; Gopinath et al., 2025; Bonadio et al., 2025). Our point of departure is that realignment extends beyond trade flows to multinational production (MP)—and, with it, the technology diffusion that MP facilitates (Ramondo and Rodríguez-Clare, 2013; Keller and Yeaple, 2013; Cai and Xiang, 2025; Lu et al., 2025).

Two examples illustrate the importance of this channel. The first concerns Samsung's presence in China. Samsung established its first major manufacturing plant in Tianjin in 1993 and expanded production across the country for over two decades. During this period, Chinese suppliers absorbed knowledge in quality assurance, supplier auditing, electrostatic discharge control, and supply-chain traceability—capabilities transferred through sustained interaction with a multinational producer. Beginning in 2018, amid escalating U.S.–China geopolitical tensions, Samsung accelerated its withdrawal; by 2020, its last LCD facility in China had closed. Production capacity shifted to Vietnam, where Samsung now assembles devices and components at scale around Hanoi. This was not merely a relocation of factories. It was a redirection of knowledge flows: the supplier development and process-control spillovers once accruing to Chinese firms are now building capabilities across Vietnam's electronics ecosystem.

The second episode highlights a distinct diffusion channel—one that operates through trade in advanced inputs rather than through local multinational production. Prior to 2022, Chinese firms acquired NVIDIA's A100 and H100 accelerators and, with them, the full accompanying knowledge bundle: the CUDA software stack, drivers, libraries, and developer support. Learning occurred through documentation and SDKs, reference designs, benchmarking, teardowns, and targeted reverse engineering of implementation details. U.S. export controls imposed in October 2022 severed this import channel; by October 2025, NVIDIA's data-center GPU market share in China had collapsed from roughly 95 percent to near zero. Fragmentation, in this case, didn't just redirect diffusion. It halted it.

These episodes point to a broader analytical challenge. Realignment simultaneously reshapes where multinationals produce, what countries trade, and how knowledge diffuses across borders—yet existing quantitative frameworks have largely treated these margins in isolation. In this paper, we embed both multinational production and technology diffusion into a unified quantitative framework to evaluate how global fragmentation

affects long-run growth and welfare. We measure how alternative realignment paths reallocate production, rewire knowledge flows, and ultimately shape incomes across countries.

We proceed in three steps. First, we lay out a semi-endogenous growth model of multinational production that features both tariff and non-tariff components of trade costs, and that incorporates subnational regions within the United States. Second, combining gravity equations with bilateral MP and trade flow data from 2000 to 2019, we recover changes in MP costs and trade costs for each country pair. Third, we quantify the welfare implications of this dual fragmentation through a series of counterfactual exercises. We evaluate the overall welfare effects of observed cost changes across countries and US states, decompose these effects into contributions from MP versus trade costs and from static reallocation versus endogenous technological change, and isolate the differential welfare exposure of third countries and US states to cost changes linked specifically to the US and to China. Using the recovered cost changes, we then classify countries into integration groups: those experiencing declining MP (trade) costs with the US but rising MP (trade) costs with China join the US MP (trade) integration group, and vice versa. We document the divergence between MP and trade integration patterns and examine the persistence of group membership across periods. Finally, we conduct counterfactual exercises that reassign economies to alternative integration groups, asking whether aligning trade and MP integration patterns improves welfare, and apply this framework to the subnational reorientation of California.

Building on Cai and Xiang (2025), we construct a dynamic multi-location model where knowledge stocks and bilateral costs jointly determine global trade and production patterns. The model features MP, trade, and technology diffusion in a multi-country Ricardian world. Key extensions relative to Cai and Xiang (2025) include the incorporation of subnational geographic units, specifically US states, which allows us to study the heterogeneous effects of global fragmentation within a single country; the introduction of technology diffusion through both trade and multinational production channels, so that knowledge spillovers arise from sellers as well as producers; and the explicit decomposition of trade costs into tariff and non-tariff components, a distinction that becomes particularly relevant given the increased salience of tariff policies in recent trade disputes. Over time, knowledge stocks evolve endogenously through these diffusion channels, so that changes in trade and MP costs affect welfare not only through static reallocation but also through their dynamic impact on knowledge accumulation. Together, these features allow us to study the welfare effects of decoupling in global production and trade across countries, within countries across space, and over time.

We show how to combine the structural gravity equation with MP and trade flow data to recover bilateral costs. We first use the observed bilateral MP and trade data to solve for the unobserved trilateral trade flows within the model's framework. With these trilateral shares determined, we invert the model to obtain the model-consistent knowledge stock for each country. Finally, combining these inferred knowledge stocks with real output per worker from the data provides a complete set of constraints, allowing us to uniquely recover the full matrices of bilateral MP and trade costs that rationalize the observed data.

After recovering the costs, we document the evolution of global economic relationships from 2000 to 2019, partitioned into three periods: 2000 to 2005, 2005 to 2016, and 2016 to 2019. Two findings emerge from the data. First, trade stagnated in the aggregate and fragmented bilaterally between the US and China after 2016, with both economies redirecting trade toward other partners. Second, in contrast to trade, China's outward multinational production expanded globally throughout 2005 to 2019, including into the US, even as bilateral trade tensions intensified.

We then quantify the welfare and growth implications of dual fragmentation through a series of counterfactual exercises, focusing on the 2016 to 2019 period. We begin by evaluating how observed changes in trade and MP costs affected welfare outcomes. By comparing 2019 outcomes to a counterfactual where costs remain fixed at 2016 levels while allowing knowledge stocks to evolve endogenously, we find that the median country experienced a real income gain of approximately 0.79%, with large heterogeneity across countries. Decomposing this effect, we find that MP cost changes account for the dominant share of welfare variation for the majority of economies. We further show that endogenous knowledge accumulation amplifies welfare gains substantially relative to a purely static analysis. At the subnational level, we document significant variation across US states, where, in contrast to the cross-country analysis, trade cost changes exert a larger effect than MP cost changes. Cross-state heterogeneity in welfare outcomes persists even when all states face identical aggregate cost shocks, confirming that differences in underlying economic structure, not only differences in cost changes, drive the uneven distribution of gains and losses.

We next isolate the differential exposure of third countries and US states to cost changes linked to the US and to China. Changes in bilateral costs with the US generate welfare effects ranging from -1.08% to 4.53% across economies, with MP costs accounting for the dominant share. Changes in bilateral costs with China produce smaller effects, ranging from -0.63% percent to 0.98%, and are driven primarily by trade costs rather than MP costs. For US states, changes in China-linked costs prove largely detrimental, with the combined effect of trade and MP cost changes with China negative for most states.

Using changes in MP and trade costs, we classify countries into China-integrating, US-integrating, or non-integrating groups, following Bonadio et al. (2025). Our classification reveals asymmetries between production and trade-based economic alignments. In trade-based classification, most countries remain non-integrating with relatively balanced group memberships. In MP-based classification, a larger share of countries align with one of the two major powers, with China consistently attracting more MP-integrating members than the United States. Countries often do not exhibit consistent alignment across both MP and trade dimensions within the same period: an economy may be China-integrating in MP while being US-integrating or non-integrating in trade, or vice versa. The analysis also reveals fluidity in countries' positions across time periods, with fewer than half maintaining consistent alignment from one period to the next. Direct transitions between opposing groups are rare in MP-based classification but more common in trade, suggesting that MP networks exhibit stronger path dependencies than trading relationships.

Given the prevalence of split alignments, we examine counterfactual scenarios where countries achieve full consolidation in both trade and MP dimensions with either the US or China integration group. The results show that consolidation would predominantly reduce welfare. The majority of economies exhibit both negative static gains and negative welfare effects from forcing consistent integration, indicating that the current pattern of misaligned integration appears economically rational. For economies switching to full China integration, welfare effects are generally smaller than static effects, reflecting negative growth effects through disrupted technology diffusion. More broadly, a comparison between static gains and welfare effects reveals that consolidation can generate positive short-run efficiency improvements that are offset, and even reversed, by impaired knowledge accumulation in the long run. When only the tariff component of trade costs is adjusted, welfare effects diminish by an order of magnitude, indicating that non-tariff components such as transportation costs, non-tariff barriers, and geopolitical frictions drive most welfare implications of integration group alignment (Fan et al., 2025). We also apply this framework to subnational reorientation, examining a counterfactual in which California shifts to the China-integrating group, and find a modest positive welfare gain driven primarily by trade integration with China.

This paper contributes to three strands of literature. First, we extend the growing body of work on fragmentation and the slowdown in globalization (Antràs, 2020), including analyses of Sino-US tariff frictions (Amiti et al., 2019; Fajgelbaum et al., 2024) and Brexit (Sampson, 2017; Dhingra and Sampson, 2022). Gopinath et al. (2025) and Bonadio et al. (2025) find that trade costs decline within blocs and increase between blocs, documenting fragmentation in trade. While existing studies focus primarily on trade disintegration

through tariffs and non-tariff barriers, we complement these papers by analyzing global production decoupling through multinational networks. Recent work examines how fragmentation affects foreign direct investment flows (Alfaro and Chor, 2023; Aiyar et al., 2024; Pellegrino et al., 2025), but we go further by quantifying the macro implications for growth and welfare through both static efficiency losses and dynamic technology diffusion effects.

Second, this paper contributes to the burgeoning literature on geoeconomics and trade (Alesina et al., 2000; Caliendo and Parro, 2015; Caliendo et al., 2019; Caliendo and Parro, 2020; Caliendo et al., 2021; Couttenier et al., 2024; Becko and O'Connor, 2024; Attinasi et al., 2025; Abeliansky and Mayrhuber, 2025; Qiu et al., 2025; Mayer et al., 2025; Clayton et al., 2025a,b; Fan et al., 2025; Ignatenko et al., 2025; Rodríguez-Clare et al., 2025; Xue, 2026).[1] While this literature emphasizes how states use economic tools for strategic goals, we provide a framework to quantify the economic consequences when countries simultaneously weaponize both trade and multinational production relationships. The most closely related empirical work is Bonadio et al. (2025), who measure trade fragmentation through data-driven changes in U.S. and China trade costs from 2015–2023. We complement their work by introducing multinational production integration groups, which prove crucial for understanding the growth and welfare implications of global fragmentation.

Third, we extend Cai and Xiang (2025) by incorporating tariffs and diffusion through both trade and MP. This builds on foundational work on multinational production (Ramondo and Rodríguez-Clare, 2013; Tintelnot, 2017; Arkolakis et al., 2018; Alviarez et al., 2023; Li, 2023; Xiang, 2023; Fan, 2025; Cai and Xiang, 2025; Lu et al., 2025) and technology diffusion (Kortum, 1997; Eaton and Kortum, 1999; Lucas, 2009; Alvarez et al., 2013; Buera and Oberfield, 2020; Perla et al., 2021; Benhabib et al., 2021; Lind and Ramondo, 2022; Cai et al., 2022, 2025). Our contribution is to show how multinational production and technology diffusion that it facilitates shape the consequences of global fragmentation and decoupling.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 describes the data and empirical strategy for backing out cost changes. Section 4 presents economic fragmentation patterns. Section 5 reports welfare effects of actual cost changes observed in data. Section 6 examines trade and MP integration patterns for different countries. Section 7 concludes.

[1] Early work dates back to Hirschman (1945). See Mohr and Trebesch (2025) for a comprehensive review.

2 Model

In this section we outline the model we use, which follows Cai and Xiang (2025). The main difference in this paper is that we explicitly model both tariff and non-tariff component of trade costs, a distinction that becomes particularly relevant given the increased salience of tariff policies in recent trade disputes. We also incorporate subnational regions within the United States and other countries into this model to study the heterogeneous effects of global fragmentation within a single country and across countries.

Basic Environment. The model features MP, trade and technology diffusion in a multi-location Ricardian world. Consider a world economy with N countries where labor is the only factor of production. In each country there is a continuum of goods of measure one. The production function for each variety $v \in [0, 1]$ has constant returns to scale, i.e., $y(v) = ql(v)$, where $y(v)$ is output, q is productivity, and $l(v)$ is labor input.

There is a large set of potential firms with different productivity to produce each variety. A firm originated in h (the *headquarter* country) is fully described by a vector of productivity $\mathbf{q}_h(v) = (q_{1h}(v), q_{2h}(v), \ldots, q_{Nh}(v))$, where $q_{jh}(v)$ is the productivity if the firm were to set up a plant in country j. The firm suffers an efficiency loss in producing abroad in the form of an iceberg MP cost $\gamma_{jh} \geq 1$, with $\gamma_{jj} = 1$. When the firm produces in j, its productivity is $q_{jh}(v)/\gamma_{jh}$.

When a firm serves market i with goods produced in j, it incurs trade cost $\tau_{ij} \geq 1$, with $\tau_{ii} = 1$. We assume that trade barriers obey the triangle inequality, i.e., $\tau_{in} \leq \tau_{ij}\tau_{jn}$ for any three countries i, j, n. In particular, $\tau_{ij} = (1 + t_{ij})(1 + d_{ij})$, where t_{ij} is the tariff rate and d_{ij} is the iceberg transportation cost.

Consumers all over the world have the same preferences over the continuum of varieties, which are constant elasticity of substitution (CES) with elasticity $\sigma > 1$. Consider consumers in country i. They get utility $u(C_{i,t})$ from the CES aggregate $C_{i,t} = \left(\int_0^1 c_{i,t}(v)^{\frac{\sigma-1}{\sigma}} dv\right)^{\frac{\sigma}{\sigma-1}}$ where $c_{i,t}(v)$ is the consumption of variety v. The associated price index is $P_{i,t} = \left(\int_0^1 p_{i,t}(v)^{1-\sigma} dv\right)^{1/(1-\sigma)}$, where $p_{i,t}(v)$ is the price of variety v in country i.

MP and Trade. Consider a firm whose headquarter is in h that produces variety v in country j. The efficiency level, $q_{jh}(v)$, is discounted by the MP cost, γ_{jh}, so the unit cost of production is $w_j\gamma_{jh}/q_{jh}(v)$, where w_j is the wage rate in country j. If the firm ships the good to market i, the unit cost incorporates transportation cost, τ_{ij}, as well. Denoting this augmented cost as $\kappa_{ijh,t} = \tau_{ij,t}w_{j,t}\gamma_{jh,t}$, the unit cost of serving market i with goods produced in j using ideas from h is $p_{ijh,t}(v) = \kappa_{ijh,t}/q_{jh}(v)$.

Consumers take prices as given. We assume perfect competition so $p_i(v)$ is equal to the lowest cost to serve market i:

$$p_{i,t}(v) = \min\{p_{ijh,t}(v);\ j,h = 1,\ldots,N\}.$$

New Ideas and Technology Frontier. Now we describe the dynamics of new ideas.[2] In each country h, new ideas arrive as a Poisson process. Each idea is about techniques to produce a single variety, drawn randomly from the uniform distribution on $[0,1]$. An idea can be specified by a random vector of efficiencies of production in all countries,

$$\boldsymbol{Q} = \boldsymbol{Z} \cdot \left(Q_p^{'}\right)^{\rho_1} \left(Q_s^{'}\right)^{\rho_2},$$

The original component, random vector $\mathbf{Z}$, is drawn from an exogenous distribution. There are two insight components, where $Q_p^{'}$ is the quality of an new idea drawn from producers, and $Q_s^{'}$ is the quality of an new idea drawn from sellers. The parameter ρ_1 and ρ_2 govern the learning intensities.

In country h, the arrival rate of ideas with original component $\mathbf{Z}$ better than $\mathbf{z}$ in at least one dimension is $A_{h,t}(\mathbf{z}) = \alpha_{h,t}\left(\sum_{n=1}^{N} z_n^{-\frac{\theta}{1-\eta}}\right)^{1-\eta}$ for any $\mathbf{z} \in \mathbb{R}_+^N$ and $\mathbf{z} \neq \mathbf{0}$, where $\alpha_{h,t}$ is an exogenous arrival intensity at time t. The parameter $\theta > 0$ determines the shape of the right tail. The parameter $\eta \in [0,1)$ measures the correlation among entries of vector $\mathbf{Z}$.

The insight component, Q_p', is drawn randomly from the distribution of the efficiency level of all producers in country h regardless of the variety they produce, and Q_s' is drawn randomly from the distribution of the efficiency level of all sellers to country h. We denote the distribution of efficiencies of actively producing plants located in country h at time t as $G_{h,t}(q')$, and the distribution of efficiencies of sellers to country h as $H_{h,t}(q')$.

The state of the art for producing a particular variety v using ideas originated in h is a vector $\mathbf{Q}^{(1)}(v)$, where the n-th entry in the vector is the highest efficiency of producing that variety in country n. Because of randomness in the arrival and quality of new ideas in h, the vector is random, with a cumulative distribution function $F_{h,t}$ characterized as a multivariate Fréchet distribution,

$$\Pr[\mathbf{Q}^{(1)}(v) \leq \mathbf{q}] = F_{h,t}(\mathbf{q}) = \exp\left(-\lambda_{h,t}\left(\sum_{n=1}^{N} q_n^{-\frac{\theta}{1-\eta}}\right)^{1-\eta}\right),$$

[2]We introduce multi-dimensional ideas to Buera and Oberfield (2020) to incorporate the model of MP.

with support $q_n > 0$ for all n.[3] The location parameter

$$\lambda_{h,t} := \int_{-\infty}^{t} \alpha_{h,\tau} \int_0^\infty \int_0^\infty x^{\rho_1\theta} dG_{h,\tau}(x) \;\; y^{\rho_2\theta} dH_{h,\tau}(y) d\tau$$

is the cumulated knowledge stock of country h at time t. The derivations are provided in Appendix A.1.

General Equilibrium. Given the results of the technology frontier, following Ramondo and Rodríguez-Clare (2013), we can characterize the allocation of expenditures and production with only the knowledge stocks, λ, and the augmented cost bundles, κ. In particular, the share of total expenditure in market i on goods produced using technology from h (the expenditure share) is

$$\pi^E_{ih,t} = \frac{\lambda_{h,t} \left(\sum_{k=1}^N \kappa_{ikh,t}^{-\frac{\theta}{1-\eta}} \right)^{1-\eta}}{\sum_{n=1}^N \lambda_{n,t} \left(\sum_{k=1}^N \kappa_{ikn,t}^{-\frac{\theta}{1-\eta}} \right)^{1-\eta}}.$$

The goods i buys from h will be produced in the countries where the cost is the lowest. The conditional probability that such a good is produced in j is

$$\psi_{ijh,t} = \frac{\kappa_{ijh,t}^{-\frac{\theta}{1-\eta}}}{\sum_{k=1}^N \kappa_{ikh,t}^{-\frac{\theta}{1-\eta}}}.$$

It follows that the share of expenditure of i on goods produced in j and originating in h (trilateral share) is

$$\pi_{ijh,t} = \psi_{ijh,t} \cdot \pi^E_{ih,t} = \frac{\kappa_{ijh,t}^{-\frac{\theta}{1-\eta}}}{\sum_{k=1}^N \kappa_{ikh,t}^{-\frac{\theta}{1-\eta}}} \frac{\lambda_{h,t} \left(\sum_{k=1}^N \kappa_{ikh,t}^{-\frac{\theta}{1-\eta}} \right)^{1-\eta}}{\sum_{n=1}^N \lambda_{n,t} \left(\sum_{k=1}^N \kappa_{ikn,t}^{-\frac{\theta}{1-\eta}} \right)^{1-\eta}}.$$

We can easily construct other aggregate shares of interest from the trilateral shares. For example, the share of total expenditure in market i on goods produced in j (the bilateral trade share) is $\pi^T_{ij} = \sum_{h=1}^N \pi_{ijh}$.

We define the equilibrium in the static model as follows. At each period t, given the knowledge stocks $\{\lambda_{i,t}\}_{i=1}^N$, MP costs, tariff rates and trade costs $\{\gamma_{ij,t}, t_{ij,t}, d_{ij,t}\}_{i=1,j=1}^{N,N}$, and

[3] Here, $\mathbf{Q}^{(1)}(v) \leq \mathbf{q}$ means $Q_n^{(1)}(v) \leq q_n$ for all $n \in \{1, \ldots, N\}$.

labor $\{L_{i,t}\}_{i=1}^N$, a *temporary equilibrium* is a vector of wages $\{w_{i,t}\}_{i=1}^N$ that satisfies the labor market clearing condition for all economies

$$w_{nt}L_n = \sum_{i=1}^{N} \frac{1}{1+t_{in,t}} \pi_{in}^T X_{it},$$

where the total expenditure X_{nt} includes lump sum transfer from tariff revenue

$$X_{nt} = w_{nt}L_n + \sum_{i=1}^{N} \frac{t_{ni,t}}{1+t_{ni,t}} \pi_{ni}^T X_{nt}.$$

Given the production and trade pattern in the temporary equilibrium, the law of motion for knowledge stock is

$$\dot{\lambda}_{h,t} = \alpha_{h,t}\Gamma_{\rho_1,\rho_2} \left[\sum_n \frac{\pi_{hhn,t}}{\sum_n \pi_{hhn,t}} \left(\frac{\lambda_{n,t}}{\pi_{hn,t}^E \psi_{hhn,t}^{1-\eta}} \right)^{\rho_1} \right] \left[\sum_n \sum_j \frac{\pi_{hjn,t}}{\sum_j \sum_n \pi_{hjn,t}} \left(\frac{\lambda_{n,t}}{\pi_{hn,t}^E \psi_{hjn,t}^{1-\eta}} \right)^{\rho_2} \right]$$

for $h = 1, \ldots, N$, where $\Gamma_{\rho_1,\rho_2} = \Gamma(1-\rho_1) \cdot \Gamma(1-\rho_2)$. The derivations are provided in Appendix A.2.

Given initial knowledge stocks $\{\lambda_{i,0}\}_{i=1}^N$, the path of arrival rates, costs, and labor allocations $\left(\{\alpha_{h,t}\}, \{\gamma_{ij,t}, \tau_{ij,t}\}_{i=1,j=1}^{N,N}, \{L_{i,t}\}_{i=1}^N\right)_{t\geq 0}$, a *sequential equilibrium* that describes the whole path of the world economy is a sequence of wage rates and knowledge stocks $\left(\{w_{i,t}\}_{i=1}^N, \{\lambda_{i,t}\}_{i=1}^N\right)_{t\geq 0}$ that at each point in time, (a) $\{w_{i,t}\}_{i=1}^N$ is the temporary equilibrium given $\{\lambda_{i,t}\}_{i=1}^N$ and $\{\gamma_{ij,t}, t_{ij,t}, d_{ij,t}\}_{i=1,j=1}^{N,N}$; (b) $\{\lambda_{i,t}\}_{i=1}^N$ follows the law of motion specified above.

If labor, trade cost and MP cost do not change in the long run, and if arrival rate of ideas has a common constant growth rate: $\dot{\alpha}_{h,t}/\alpha_{h,t} = g_\alpha$, a balanced growth path exists. Along the balanced growth path all equilibrium variables grow at a constant rate or stay constant.

3 Data and Calibration

To quantify the welfare effects of cost changes and decoupling, we need to measure changes in bilateral MP and trade costs, which requires data on both trade flows and MP activities across locations. Our quantitative model has 50 U.S. states and 75 other economies. This section presents the data sources employed and describes how we recover MP and trade costs from observed trade and MP flows in a manner consistent with our

theoretical framework.

3.1 Data

For bilateral tariff rates, we use the Global Tariff Database from Teti (2024), which provides trade-weighted average bilateral tariffs and unweighted average bilateral tariffs. This allows us to construct the $N \times N$ tariff matrix between all importer-exporter pairs in our sample.[4] For trade data, we use trade flows from the OECD Inter-Country Input-Output tables (ICIO 2025 release) (OECD, 2025), which provide flows of international trade in goods and services between countries measured in basic prices (Yamano et al., 2023).[5]

Bilateral trade flows among the 50 U.S. states are constructed by combining information from the ICIO and the Commodity Flow Survey (CFS).[6] From the ICIO, we get total U.S. domestic sales for each year. From the CFS, we derive bilateral expenditure shares across states. These two components allow us to construct bilateral trade flows for each U.S. state, ensuring that total U.S. domestic sales align with the ICIO data for each year.

Bilateral trade flows between the 50 states and other economies in our sample are constructed by combining information from the ICIO with employment data across states and sectors from the Bureau of Economic Analysis (BEA). We allocate the country-level bilateral trade flows from the ICIO to the state level by assuming that the share of each state in total U.S. trade with any country in each sector is determined by the state share of total employment in that sector (Caliendo et al., 2019). Aggregating these bilateral trade flows across sectors yields the bilateral trade flows between the 50 U.S. states and other countries. Hence, we are able to construct the $N \times N$ matrix of trade shares, π_{ij}^{T}.

To derive the empirical counterpart of bilateral MP data in the model, we use the Analytical Activities of MNEs (AMNE) database from OECD.[7] The bilateral output matrix breaks down global gross output by producer country, industry, and country of ownership,

[4]Specifically, the data source is Feodora Teti's Global Tariff Database (Version Beta1-2024-12). This database provides bilateral tariff rates for 200 importers and their partners, spanning 34 years (1988-2021). Its primary data are derived from multiple authoritative sources, including the WTO's RTA Database, the WTO's Integrated Data Base (IDB), ITC's Market Access Map, and US and EU tariff schedules from national authorities. Trade-weighted tariffs are constructed using contemporaneous trade data from CEPII's BACI starting from 1995. For details on the database construction, see Teti (2024).

[5]The OECD-ICIO Tables are presented for 80 economies and the Rest of the World from 1995 to 2022. For details on the database construction, see Yamano et al. (2023).

[6]The Commodity Flow Survey (CFS), is conducted every five years for the Bureau of Transportation Statistics (BTS) by the U.S. Census Bureau (Census) as part of the Economic Census, which is a shipper survey of approximately 165,000 establishments and provides a comprehensive, multimodal picture of in-scope national freight flows and represents the only publicly available source of data for highway freight.

[7]The Analytical AMNE database includes 41 industries and 76 economies and the rest of the world region from 2000 to 2020. For details on the database construction, see Cadestin et al. (2018).

which allows us to construct the MP shares, π^M_{jh} between countries.

Bilateral MP flows between the 50 U.S. states and other countries are constructed by combining data from the AMNE and state-sector employment from BEA, following the same approach used for trade flows. Specifically, we allocate country-level bilateral MP flows in each sector from the AMNE to the states by in proportion to each state's share of total U.S. employment in the relevant sector. Aggregating these sector-level bilateral MP flows across sectors yields state-level bilateral MP flows with each foreign economy.

For bilateral MP flows across U.S. states, we assume that each state's production share in total U.S. MP activity of each sector is proportional to the state's share of total employment in that sector. Thus, we can derive the total production in each U.S. state. We will describe how MP flow among U.S. states is obtained in the next part.

Finally, real national accounts data, including real GDP (X_i/P_i) and employment (emp_i), for individual countries come from the Penn World Table 10.01 (Feenstra et al., 2015), which covers the period from 1950 to 2019. For U.S. states, we combine state-level real GDP (X_i/P_i) and employment (emp_i) data from BEA with U.S. aggregate data from the Penn World Table. We use the BEA data to get each state's share of the national total to split the U.S. aggregation from the Penn World Table.

3.2 Calibration

We first take parameters off-the-shelf from Cai and Xiang (2025), setting $\theta = 4.5$, $\eta = 0.55$, $\beta = 0.95$, $\rho_1 = 0.629$, $\rho_2 = 0.488$. Next, since bilateral MP flows between U.S. states are not directly observed, we impose a simplifying assumption: the bilateral MP cost is identical across all distinct pairs of U.S. states, denoted by γ_{state}. Formally, if we denote the set of states as $\mathcal{S}$, the set of economies as $\mathcal{C}$,

$$\gamma_{s_1,s_2} = \gamma_{\text{state}}, \quad \forall\, s_1, s_2 \in \mathcal{S},\ s_1 \neq s_2. \tag{1}$$

Having described the data and assumptions, we now turn to the calibration procedure. The goal is to recover the N^3 trilateral shares π_{ijh}, the knowledge stock λ_n, the bilateral trade costs τ_{ij}, and the bilateral MP costs γ_{jh}.

We construct the trilateral shares based on the bilateral trade shares $\pi^T_{ij,t}$ and the bilateral MP shares $\pi^M_{jh,t}$. Define:

$$\xi^M_{jh,t} = \lambda_{h,t}\gamma^{-\theta}_{jh,t}, \qquad \xi^T_{ij,t} = (\tau_{ij,t}w_{j,t})^{-\theta}, \tag{2}$$

We can rewrite the trilateral share equation as:

$$\pi_{ijh,t} = \frac{(\xi^T_{ij,t}\xi^M_{jh,t})^{1/(1-\eta)}}{\sum_{k=1}^N (\xi^T_{ik,t}\xi^M_{kh,t})^{1/(1-\eta)}} \frac{\left(\sum_{k=1}^N (\xi^T_{ik,t}\xi^M_{kh,t})^{1/(1-\eta)}\right)^{1-\eta}}{\sum_{n=1}^N \left(\sum_{k=1}^N (\xi^T_{ik,t}\xi^M_{kn,t})^{1/(1-\eta)}\right)^{1-\eta}}. \tag{3}$$

Let $X_{i,t}$ denote total expenditure in i, and $Y_{j,t} = \sum_i \sum_h X_{i,t}\pi_{ijh,t}\frac{1}{1+t_{ij,t}}$ total production in j. By definition, the bilateral trade share is $\pi^T_{ij,t} = \sum_h \pi_{ijh,t}$, and the bilateral MP share is $\pi^M_{jh,t} = \sum_{i,t} X_{i,t}\pi_{ijh,t}\frac{1}{1+t_{ij,t}} / \sum_i \sum_h X_{i,t}\pi_{ijh,t}\frac{1}{1+t_{ij,t}}$. Substituting yields:

$$\pi^T_{ij,t} = \sum_{h=1}^N \frac{(\xi^T_{ij,t}\xi^M_{jh,t})^{1/(1-\eta)}}{\sum_{k=1}^N (\xi^T_{ik,t}\xi^M_{kh,t})^{1/(1-\eta)}} \frac{\left(\sum_{k=1}^N (\xi^T_{ik,t}\xi^M_{kh,t})^{1/(1-\eta)}\right)^{1-\eta}}{\sum_{n=1}^N \left(\sum_{k=1}^N (\xi^T_{ik,t}\xi^M_{kn,t})^{1/(1-\eta)}\right)^{1-\eta}}, \tag{4}$$

$$\pi^M_{jh,t} = \sum_{i=1}^N \frac{X_{i,t}}{Y_{j,t}} \frac{1}{1+t_{ij,t}} \frac{(\xi^T_{ij,t}\xi^M_{jh,t})^{1/(1-\eta)}}{\sum_{k=1}^N (\xi^T_{ik,t}\xi^M_{kh,t})^{1/(1-\eta)}} \frac{\left(\sum_{k=1}^N (\xi^T_{ik,t}\xi^M_{kh,t})^{1/(1-\eta)}\right)^{1-\eta}}{\sum_{n=1}^N \left(\sum_{k=1}^N (\xi^T_{ik,t}\xi^M_{kn,t})^{1/(1-\eta)}\right)^{1-\eta}}. \tag{5}$$

Knowledge stocks can be recovered from the trilateral shares and the observed real output per worker which we take from the PWT, formally:

$$\lambda_{i,t} = B^{-\theta}\omega^{\theta}_{i,t}\pi_{iii,t}\psi^{-\eta}_{iii,t}, \tag{6}$$

where $B = \Gamma(1 + (1-\sigma)/\theta)^{1/(1-\sigma)}$ is a constant.

Denote the collection of observed variables by $\mathcal{D}$, which consists of:

$$\mathcal{D} = \Big\{ \{\pi^T_{ij}\}_{i,j\in\mathcal{S}\cup\mathcal{C}},\ \{\pi^M_{jh}\}_{j,h\in\mathcal{S}\cup\mathcal{C},\ (j,h)\notin\mathcal{S}\times\mathcal{S}}, \{X_i\}_{i\in\mathcal{S}\cup\mathcal{C}},\ \{Y_i\}_{i\in\mathcal{S}\cup\mathcal{C}},\ \{w_i\}_{i\in\mathcal{S}\cup\mathcal{C}},\ \{\omega_i\}_{i\in\mathcal{S}\cup\mathcal{C}},\ \{\tau_{ij}\}_{i,j\in\mathcal{S}\cup\mathcal{C}} \Big\}.$$

Moreover, we normalize the self trade τ_{ii} and self MP cost γ_{ii} to be 1. Therefore, the calibration problem reduces to recovering the structural parameters—bilateral trade costs $\{\tau_{ij}\}_{i,j\in\mathcal{S}\cup\mathcal{C}}$, MP costs $\{\gamma_{ij}\}_{i,j\in\mathcal{S}\cup\mathcal{C}}$, and knowledge stocks $\{\lambda_i\}_{i\in\mathcal{S}\cup\mathcal{C}}$—conditional on the observed dataset $\mathcal{D}$, the self trade/MP cost, and the interstate MP cost γ_{state}, subject to the system of equations given by (1)–(6). The detailed solution algorithm and the procedure to pin down γ_{state} are provided in Appendix B.

The data and parameters prepared above are employed in the quantitative exercises of Section 5. We next turn to the pattern of how trade and MP flows among major economies have evolved in recent years.

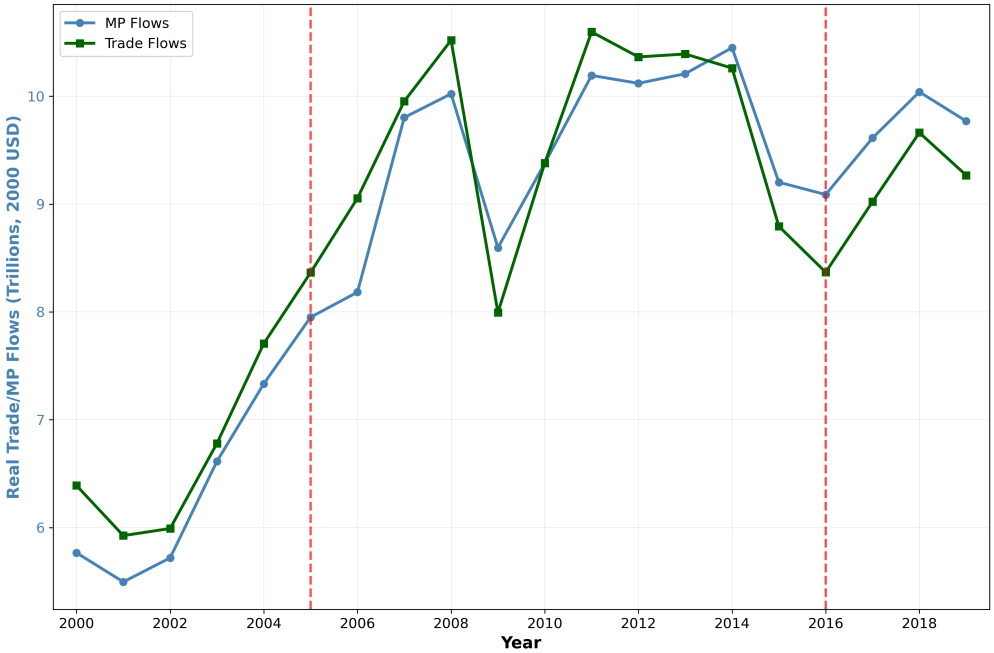


Figure 1: Evolution of MP and Trade Flows Over Time

Notes: This figure shows the time series of aggregate multinational production (MP) and trade flows between countries from 2000 to 2019. Sources: Analytical AMNE and UNCTADstat Data Center.

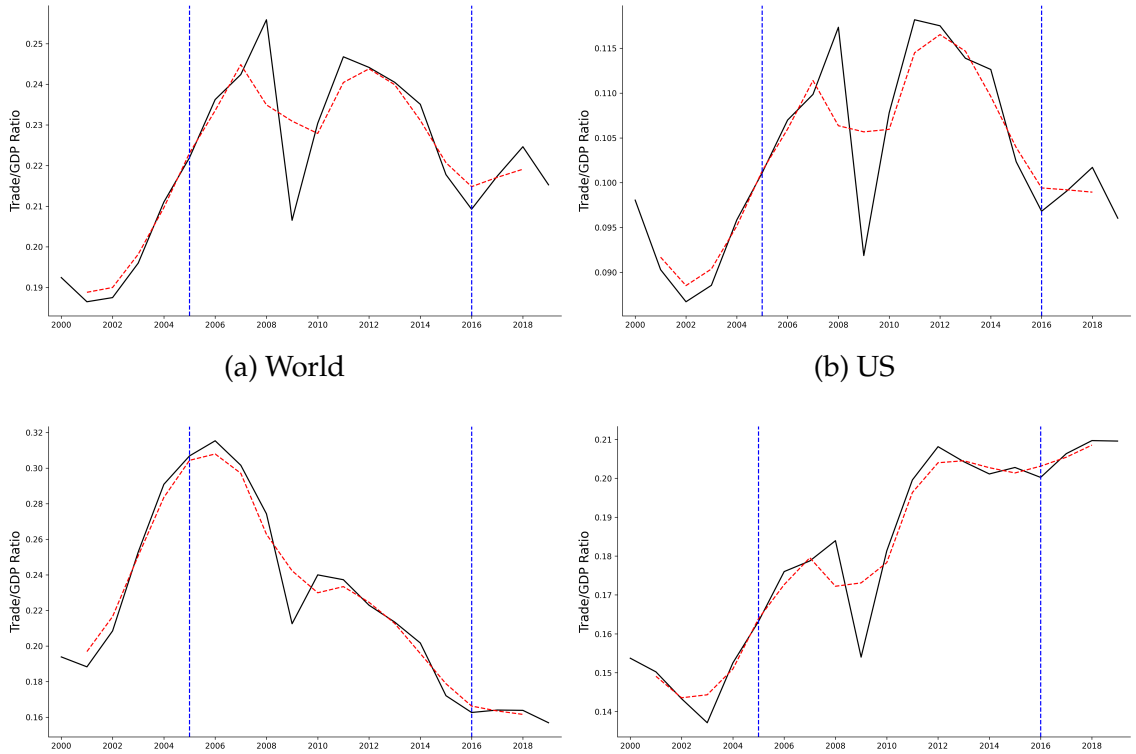
4 Global Trade Fragmentation and China's MP Expansion

Our analysis employs data spanning from 2000 to 2019, which we partition into three periods: 2000 to 2005, 2005 to 2016, and 2016 to 2019. The choice of periodization is motivated by both empirical patterns in global economic flows and key policy developments. As shown in Figure 1, both MP and trade flows experienced distinct phases of growth and adjustment that align with these periods. The middle period (2005 to 2016) corresponds to an era of multilateral trade initiatives, including the Trans-Pacific Partnership (TPP) negotiations. The final period (2016 to 2019) captures the onset of bilateral trade tensions and policy reversals.

We document two main findings. First, trade stagnated in the aggregate and fragmented bilaterally between the US and China after 2016. Second, in contrast to trade, China's multinational production expanded globally throughout 2005 to 2019, including into the US, even as bilateral trade tensions intensified.

Trade: stagnation and fragmentation. Figure 2 plots trade-to-GDP ratios from 2000 to 2019. At the global level, trade as a share of GDP stagnated after the mid-2000s. China's trade-to-GDP ratio declined substantially over this period, and the US ratio remained flat. As documented by Bonadio et al. (2025), aggregate trade-to-GDP ratios for the US, China, and the EU were largely stable after 2016.

Beneath this aggregate stability, the bilateral composition of trade shifted. Table 1 reports



(a) World (b) US

(c) China (d) EU

Figure 2: Trade / GDP Ratios

Notes: Solid line denotes the trade-GDP ratio, defined as 0.5(exports+imports)/GDP. Dashed line denotes the 3-year moving average of the trade-GDP ratio. Data Sources: DOTS and WEO.

percentage changes in bilateral import-to-GDP ratios. During 2005 to 2016 (Panel a), US imports from China increased even as US overall imports relative to GDP fell, reflecting a reorientation of US import sourcing toward China at the expense of other partners. China's imports from all regions grew more slowly than its GDP, driving the decline in its trade-to-GDP ratio. The EU saw broad-based growth in imports from all partners.

The post-2016 period reveals a different pattern: fragmentation centered on the US and China (Panel b). US imports from China declined, and China's imports from the US fell as well. Both economies redirected trade toward other partners. US imports from the EU, and from the rest of the world, increased. China's imports from the EU and the rest of the world also grew. Overall import-to-GDP ratios, however, changed little, consistent with trade diversion rather than contraction.

Multinational production: China's global expansion. Figure 3 shows the evolution of MP-to-GDP ratios. World MP grew rapidly before 2005, then moderated, and remained stable after 2016. The pattern for multinational production, however, differs from trade in an important way: China's outward MP expanded throughout the sample period, even as bilateral trade between the US and China contracted after 2016.

(a) 2005–2016

	USA	CHN	EU	ROW
USA		-0.44	0.19	-0.01
CHN	0.24		0.70	1.63
EU	-0.09	-0.43	0.07	-0.06
ROW	-0.21	-0.53	0.09	-0.08
Change in total imports-to-GDP	-0.12	-0.51	0.11	0.07

(b) 2016–2019

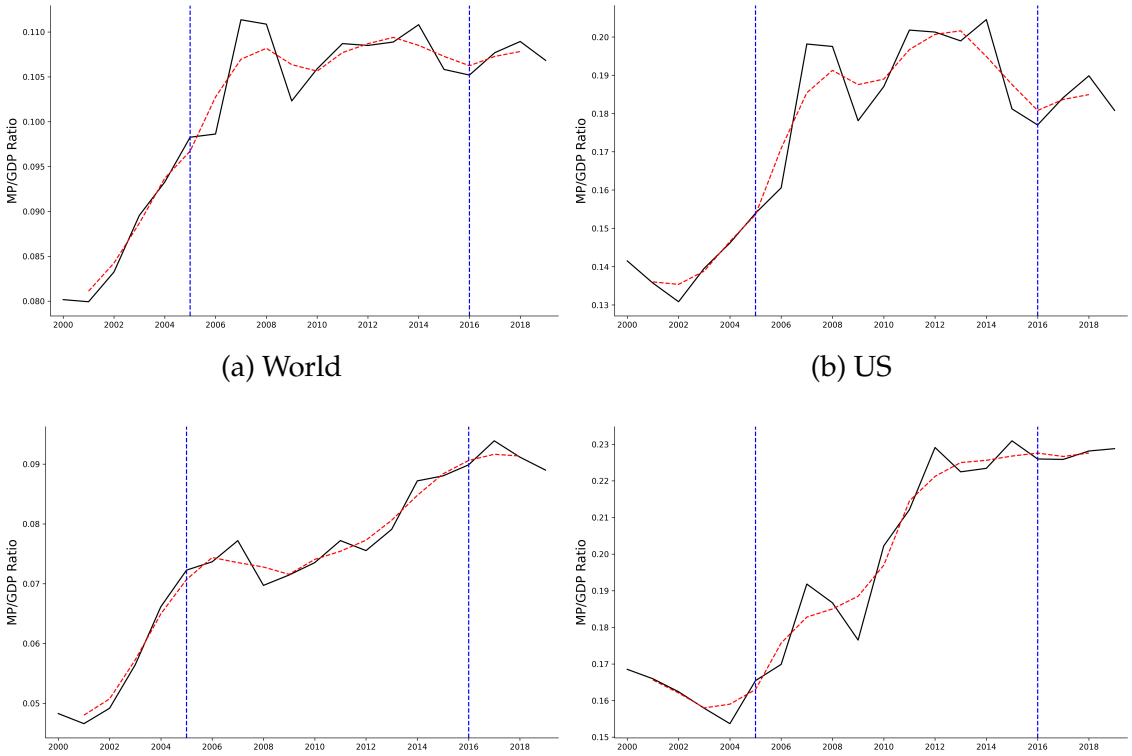
	USA	CHN	EU	ROW
USA		-0.29	0.07	0.02
CHN	-0.15		0.07	0.10
EU	0.08	0.04	0.03	0.01
ROW	0.02	0.05	0.07	0.10
Change in total imports-to-GDP	-0.01	0.02	0.05	0.06

Table 1: Changes in Trade

Notes: This table reports the percentage change in the ratio of import to GDP from the source economy (rows) to destination (columns) in different periods. Appendix C provides the ratios in the base year.

Table 2 documents this divergence. During 2005 to 2016 (Panel a), China's outward MP grew across all destinations: the US, the EU, and the rest of the world. During 2016 to 2019 (Panel b), China's outward MP continued to increase in the US, the EU, and the rest of the world, in contrast to the concurrent decline in US-China bilateral trade. The US, for its part, saw declining outward MP to both China and the EU in the post-2016 period.

The asymmetry between trade and MP in the post-2016 period is notable. While bilateral trade between the US and China fragmented, the US received increasing inward MP from China, and Chinese firms expanded their production presence in the EU and the rest of the world as well. At the same time, China's MP-to-GDP ratio declined because inward MP within China did not keep pace with China's rapid GDP growth. Taken together, these patterns suggest that China increasingly engaged in the global economy through multinational production rather than trade, extending its production presence abroad even as trade relationships became more contested.



(a) World (b) US

(c) China (d) EU

Figure 3: MP / GDP Ratios

Notes: Solid line denotes the MP-GDP ratio, defined as $0.5(\text{inward mp} + \text{outward mp})/\text{GDP}$. Dashed line denotes the 3-year moving average of the MP-GDP ratio. Data Sources: Analytical AMNE and WEO.

(a) 2005–2016

	USA	CHN	EU	ROW
USA		-0.17	0.24	-0.03
CHN	14.08		24.52	9.18
EU	0.11	0.01	0.18	-0.08
ROW	0.41	0.18	0.43	0.21
Change in total inward MP-to-GDP	0.24	0.07	0.25	0.07

(b) 2016–2019

	USA	CHN	EU	ROW
USA		-0.06	-0.01	0.04
CHN	1.11		0.51	0.38
EU	0.00	-0.06	0.01	0.00
ROW	0.09	-0.09	0.03	0.05
Change in total inward MP-to-GDP	0.05	-0.08	0.01	0.05

Table 2: Changes in MP

Notes: This table reports the percentage change in the ratio of inward MP to GDP from the source economy (rows) to destination (columns) in different periods. Appendix C provides the ratios in the base year.

5 Welfare Effects of Cost Changes

In this section, we study how the actual changes in trade and MP costs affected welfare outcomes. We focus on the 2016 to 2019 period, which, as shown in Table 1(b), captures during which bilateral trade between the US and China declined but China's outward multinational production to the US increased, while both economies simultaneously deepened their linkages with other partners. We relegate results for earlier periods to Appendix D.

Throughout this section, unless otherwise specified, we construct two scenarios to quantify the welfare effect of cost changes. In the baseline scenario, we allow trade and MP costs to evolve according to the observed data from 2016 to 2019, while knowledge stocks adjust endogenously; this replicates the actual 2019 economy, which we call the model factual in 2019.[8] In the counterfactual scenario, we hold all trade and/or MP costs fixed at their 2016 levels and let the economy evolve to 2019 with knowledge stocks again adjusting endogenously. The real income difference between these scenarios captures the welfare impact of the observed cost changes over this period.

We first quantify how observed changes in trade and MP costs affected welfare outcomes across economies. We find sizable welfare effects, with MP cost changes accounting for the dominant share of welfare variation for the majority of economies. We also find substantial effect of technological change.

These welfare effects exhibit large heterogeneity across economies. Moreover, within a single country, the effects of cost changes can vary substantially. Focusing on U.S. states, we find significant cross-state dispersion in welfare outcomes, with Connecticut, South Dakota, and North Dakota benefiting the most and West Virginia experiencing the largest losses. The heterogeneity is still present even when all states face identical aggregate cost shocks. This indicates that underlying economic structure, not just differential exposure to cost changes, drives the uneven distribution of gains and losses.

We ask how cost changes vis-à-vis the US or China alone propagated to third countries. We find that US-linked cost changes generate larger welfare effects than China-linked ones. Notably, these can yield opposite welfare implications for the same country, suggesting that economies are differentially exposed to each major power through distinct trade and MP linkages. Within the US, we also find that the effects of trade cost changes with China are mostly negative, and again the effects vary across states.

5.1 Welfare Effects of Cost Changes and Technical Change

We first quantify how observed changes in trade and MP costs during 2016 to 2019 affected welfare outcomes across economies. We report three findings. First, cost changes generate sizable welfare effects with large heterogeneity across countries. Second, MP cost changes account for the

[8] The 2019 model factual replicates 2019 real data in terms of knowledge stocks, trade flows, and MP costs, except that we abstract from trade imbalances.

dominant share of welfare variation for the majority of economies. Third, endogenous knowledge accumulation amplifies welfare gains substantially: absent technical change, median welfare gains among economies would be underestimated by 0.13 percent.

	Median (%)	p25 (%)	p75 (%)
Overall	0.791	-0.045	1.882
Trade Costs	0.192	0.053	0.543
MP Costs	0.500	-0.269	1.440

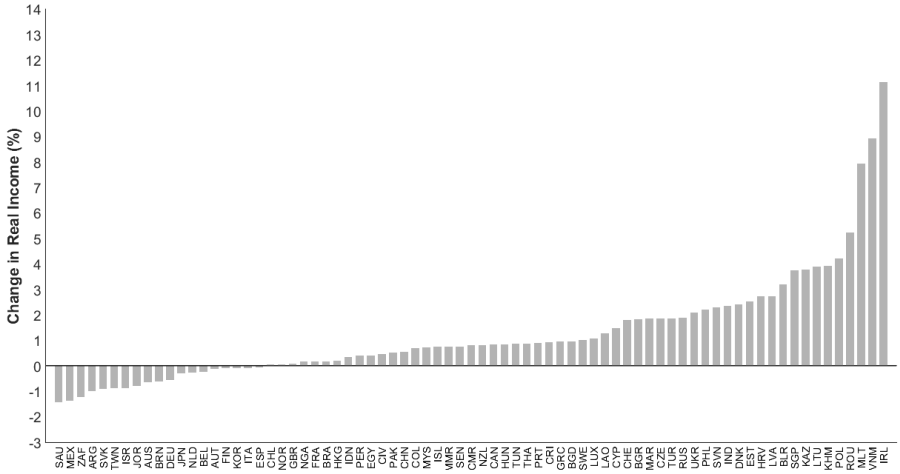
Table 3: Gains from Changes in MP and Trade Costs, 2016-2019

Notes: This table reports the cross-country distribution of percentage changes in real income resulting from observed changes in MP and trade costs between 2016 and 2019. Columns report the median, 25th percentile, and 75th percentile across countries. The model factual allows all costs and knowledge stocks to evolve to their 2019 values. "Overall" reports the real income difference between the model factual and a counterfactual holding both MP and trade costs fixed at 2016 levels. "Trade Costs" reports the difference between the model factual and a counterfactual holding only trade costs fixed at 2016 levels. "MP Costs" reports the difference between the model factual and a counterfactual holding only MP costs fixed at 2016 levels. In all scenarios, knowledge stocks adjust endogenously. Changes are calculated as (model factual − counterfactual) / model factual × 100%.

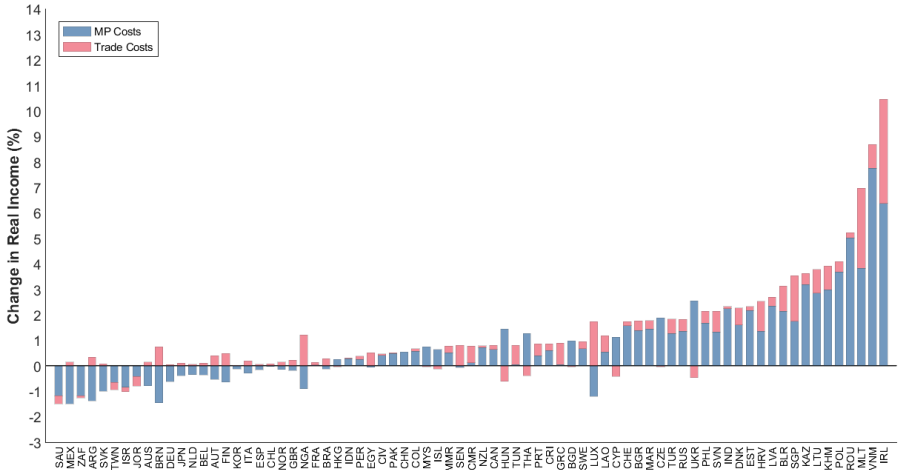
Figure 4(a) reports the results. Each bar represents the percentage change in real income attributable to cost changes for a given country. And the first row of Table 3 reports the summary statistic across economies. Two features stand out. First, the welfare effects are sizable in magnitude. For the median economy, changes in MP and trade costs in this period leads to 0.79% increase in real income, and the economy at the 75th percentile gains 1.88% in real income. Second, there is large heterogeneity across countries, both in sign and in magnitude. While Ireland and Vietnam experienced the largest gains, with 11.14% and 8.91% respectively, more then a quarter of countries in the sample experience real income loss from cost changes, among which Saudi Arabia and Mexico would have the largest loss around -1.40%.

Larger Welfare Effects from MP than Trade Cost Changes To separate the welfare impacts of MP cost changes from those of trade cost changes, we do two more counterfactual exercises. In one exercise, we hold all MP costs fixed at their 2016 levels but allow trade costs to evolve according to the observed data. The real income difference from the model factual is the welfare impact of observed MP cost changes. In the other, we fix trade costs and let MP costs to evolve as implied by data, and uncover the real income difference from the model factual as the welfare impact of observed trade cost changes.

Figure 4(b) isolates the contributions of trade and MP costs separately, each capturing full general equilibrium effects including endogenous knowledge adjustment. The height of blue bars report real income differences between the factual 2019 economy and a counterfactual holding MP



(a) Combined Effects of Changes in MP and Trade Costs



(b) Separate Effects of Changes in MP and Trade Costs

Figure 4: Gains from Trade and MP Costs Changes, 2016-2019

Notes: Panel (a) shows the combined welfare impact of trade and MP cost changes over 2016-2019 for each economy. Each bar reports the percentage change in real income between the model factual(where costs evolve according to observed data) and the counterfactual scenario (where trade and MP costs are fixed at 2016 levels). In both scenarios, knowledge stocks adjust endogenously. Changes are calculated as: (model factual − counterfactual) / model factual × 100%. Panel (b) shows welfare effects to trade costs and MP costs separately, where each attribution captures the full general equilibrium effect including induced knowledge accumulation. These two bars sum up nearly to the total welfare change shown in panel (a), with residual differences due to nonlinearities.

costs at 2016 levels. The height of red bars report the analogous comparison holding trade costs at 2016 levels. For example, in Ireland, the country with the largest impact, the welfare gains from MP cost changes is 6.37%, and from trade cost changes is 4.08%.

The central finding is that MP cost changes generate larger welfare effects than trade cost changes during this period: for the majority of economies, the blue bars exceed the red bars in absolute value and gains from MP cost changes are closer to the overall in summary statistic. Although the role of trade costs is more pronounced during 2016 to 2019 than in earlier periods (see Appendix Figure D.1 and Figure D.3), MP costs remain the primary driver of welfare variation across countries.

The Substantial Role of Technological Change Changes in costs affect welfare in two ways. Statically, they affects the trade and multinational production patterns. Lower trade costs mean access to cheaper goods. Lower MP costs mean goods can be produced with better technology, as well as higher wages for economies that receive foreign production. Both of these channels can lead to higher real income. Dynamically, changes in trade and MP also lead to changes in knowledge stocks through knowledge diffusion, which also have welfare impacts. In the previous part, we present the overall (static + dynamic) effect of cost changes. We now turn to understand how much welfare changes is from technological changes.

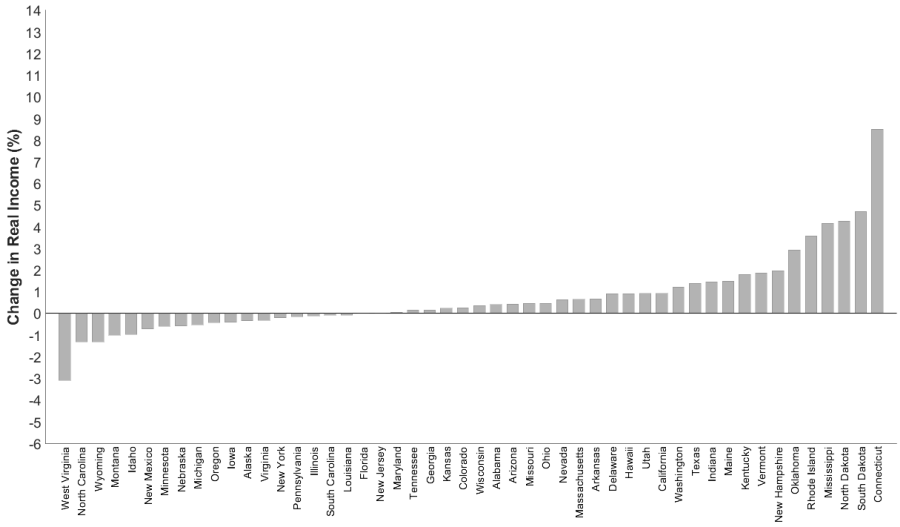
Specifically, we compute a counterfactual exercise where trade and MP costs changes from 2016 to 2019 level, but knowledge stocks are fixed at their 2016 levels. The difference of welfare in this counterfactual from the 2016 model factual is the static welfare effect of cost changes. We compare the static effect to overall effect to understand the size of dynamic effects. Knowledge stock changes contribute a sizable portion of total welfare gains for many economies. Absent technological change, the median of the static welfare effect is 0.66%, 16% lower than the 0.79% number in Table 3.

5.2 Welfare Changes Across US States

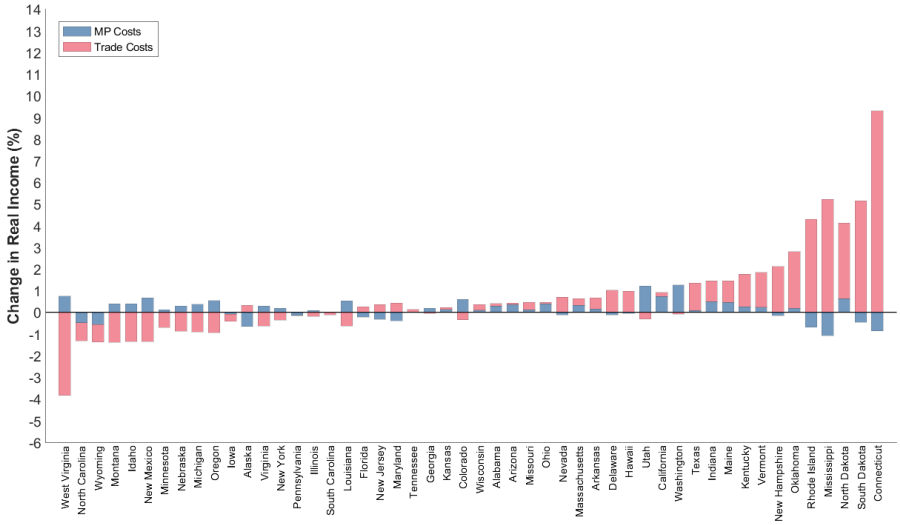
Beyond the cross-country heterogeneity documented above, we find significant variation across US states in the welfare effects of cost changes. As before, we focus on the 2016-2019 period in the main text and report results for other periods in Figures D.2 and D.4 in the appendix.

Figure 5(a) and the first row of Table 4 present the distribution of real income gains across states. Consistent with the cross-economy analysis, we find sizable gains from cost changes and substantial heterogeneity in their magnitude. The population-weighted average gain from cost changes for the United States is 0.49%. Connecticut, South Dakota, and North Dakota benefit the most from cost reductions during this period, while West Virginia experiences the largest losses.

We isolate the effects of trade and MP costs, again each including the general equilibrium effects with endogenous knowledge stock changes, as shown in Figure 5(b) and the final two row of Table



(a) Combined Effects of Changes in MP and Trade Costs



(b) Contributions of Trade and MP Cost Changes

Figure 5: Gains from Trade and MP Costs Changes, 2016-2019

Notes: Panel (a) shows the combined welfare impact of trade and MP cost changes over 2016-2019 for each state in the US. Each bar reports the percentage change in real income between the model factual (where costs evolve according to observed data) and the counterfactual scenario (where trade and MP costs are fixed at 2016 levels). In both scenarios, knowledge stocks adjust endogenously. Changes are calculated as: (model factual $-$ counterfactual) / model factual $\times$ 100%. Panel (b) shows welfare effects to trade costs and MP costs separately, where each attribution captures the full general equilibrium effect including induced knowledge accumulation. These two bars sum up nearly to the total welfare change shown in panel (a), with residual differences due to nonlinearities.

	Median (%)	p25 (%)	p75 (%)
Overall	0.297	-0.329	1.200
Trade Costs	0.127	-0.377	0.973
MP Costs	0.164	-0.118	0.385

Table 4: Gains from Changes in MP and Trade Costs, 2016-2019

Notes: This table reports the cross-state distribution of percentage changes in real income resulting from observed changes in MP and trade costs between 2016 and 2019. Columns report the median, 25th percentile, and 75th percentile across states in USA. The model factual allows all costs and knowledge stocks to evolve to their 2019 values. "Overall" reports the real income difference between the model factual and a counterfactual holding both MP and trade costs fixed at 2016 levels. "Trade Costs" reports the difference between the model factual and a counterfactual holding only trade costs fixed at 2016 levels. "MP Costs" reports the difference between the model factual and a counterfactual holding only MP costs fixed at 2016 levels. In all scenarios, knowledge stocks adjust endogenously. Changes are calculated as (model factual − counterfactual) / model factual × 100%.

4. In contrast to the cross-economy exercise, changes in trade costs exert a larger effect than changes in MP costs for US states throughout the period.

A natural question is what drives the large cross-state heterogeneity documented above. Two channels could be at work. First, states differ in their underlying economic structure, such as initial knowledge stock, MP and trade exposure, and population size, which shape how a given set of cost changes translates into welfare outcomes. Second, although tariff policy is set at the federal level and all states face the same tariff schedule, MP costs and non-tariff trade barriers may vary across states, so that states face different cost shocks to begin with. We show that cross-state heterogeneity in welfare outcomes is not solely driven by the latter: even when all states face identical aggregate cost changes, significant dispersion in welfare effects remains.

To demonstrate this, we conduct an additional exercise in which we impose identical foreign shocks to trade and MP costs on all states, calibrated to match the aggregate U.S. cost changes implied by the data over the 2016-2019 period. This design effectively shuts down any cross-state variation in cost shocks in generating heterogeneous welfare outcomes. The results, presented in Figure 6, show that substantial variation in welfare effects persists even under uniform shocks, confirming that the uneven distribution of gains and losses across states is driven not only by differences in cost changes but also by the underlying economic structure of each state.

5.3 Differential Exposure to US and China Cost Changes

Section 4 documented two key shifts in US-China economic relations during 2016-2019: bilateral trade flows between the two countries declined, and US outward multinational production to China contracted. At the same time, both economies deepened their trade and MP linkages with other partners.

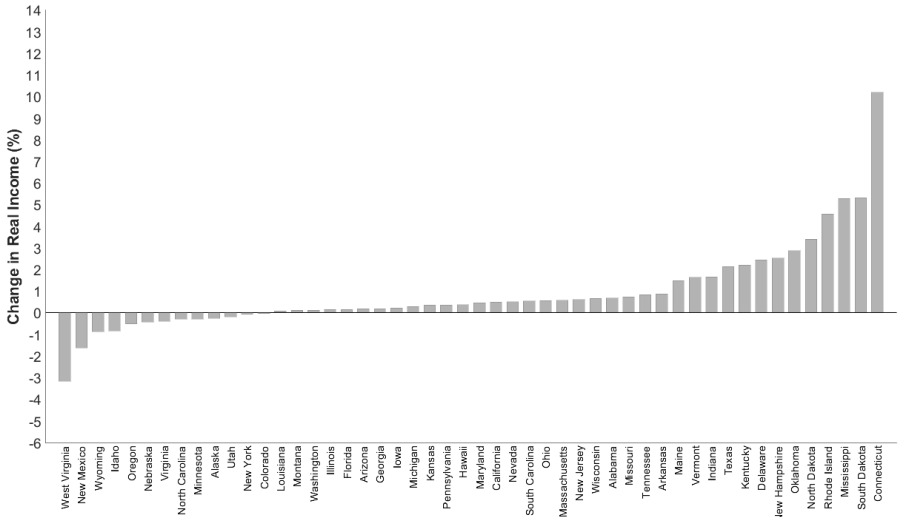


Figure 6: Gains from Common Changes in Trade and MP Costs

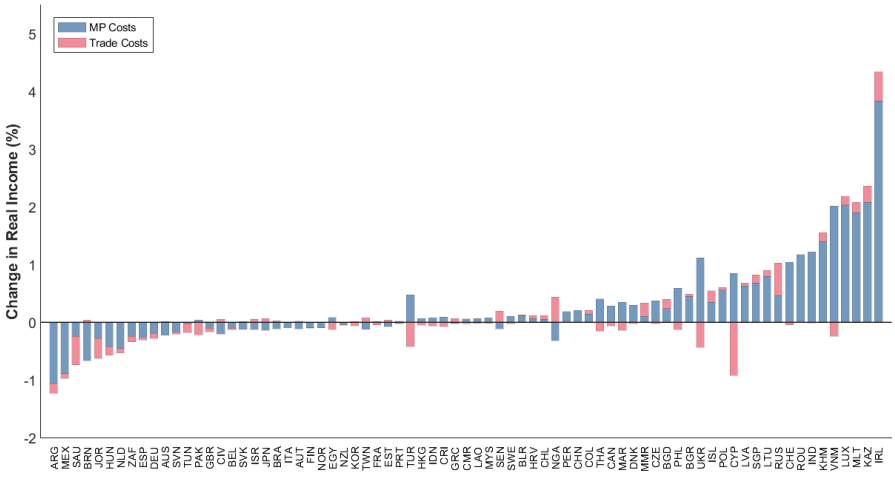
Notes: The figure shows changes in real income when trade and MP cost changes for each state is identical, equal to the overall cost changes calibrated for the United States at the country level. Each bar reports the percentage change in real income between a no-cost change counterfactual 1 (where all trade and MP costs are fixed at their 2016 levels) and a counterfactual 2 (where we impose identical foreign trade and MP costs for every U.S. state). In both scenarios, knowledge stocks evolve endogenously. The change is calculated as: (counterfactual 2 − counterfactual 1) / counterfactual 1 × 100%.

In this section, we turn to the implications of these realignments for their trading partners. Specifically, we ask how changes in the US or China costs with other partners over the 2016-2019 period affected welfare of these partners.

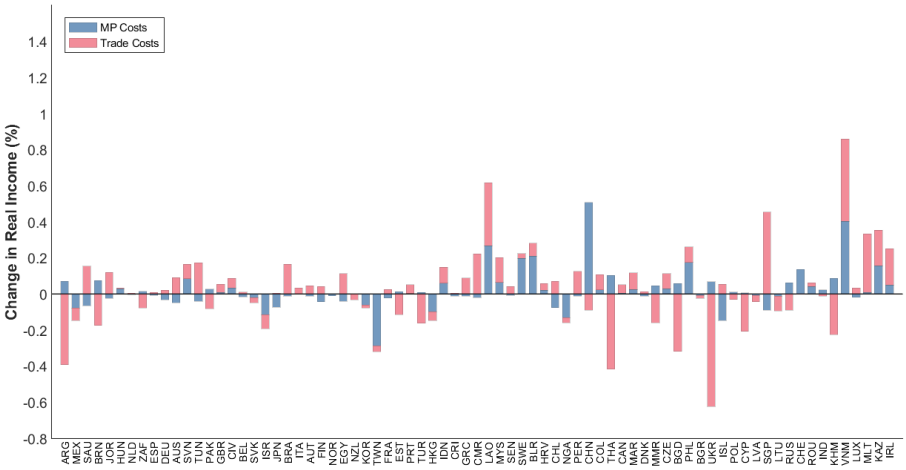
To isolate the roles of US- and China-linked cost changes, we conduct two parallel counterfactual exercises. In the first, we hold all bilateral MP and trade costs with the US fixed at their 2016 levels and allow non-US costs to evolve as observed. In the second, we perform the analogous exercise for China, holding all bilateral costs with China fixed at their 2016 levels. Comparing welfare outcomes under each counterfactual with the baseline allows us to quantify the extent to which changes in costs with the US and with China, arguably driven by endogenous policy choices, propagated to third countries.

We begin with the effects of changes in bilateral costs with the US, reported in Figure 7(a). The height of each segment represents the corresponding gain: blue segments for MP cost changes and red segments for trade cost changes. Across different economies, the welfare effects range from −1.08% to 4.53%. Changes in MP costs account for the dominant share of the welfare variation across countries. Notably, China also benefits from cost changes vis-à-vis the US, mainly driven by MP costs.

Turning to changes in bilateral costs with China, shown in Figure 7(b), the welfare effect across economies range from −0.63% to 0.98%, which are smaller in size compared to the US. However, for most economies, it is the changes in bilateral trade costs with China, rather than changes in MP



(a) Gains from Cost Changes in Bilateral Costs with the US, 2016-2019



(b) Gains from Changes in Bilateral Costs with China, 2016-2019

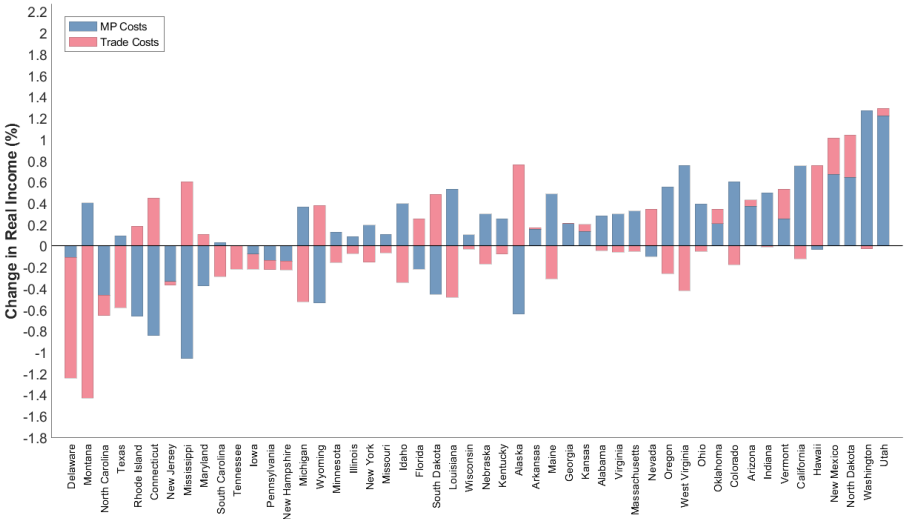
Figure 7: Gains from Changes in Bilateral Costs with the US and China

Notes: This figure shows the welfare impact of the changes in bilateral cost with U.S. and China 2016-2019 for each economy. Each bar reports the percentage change in real income between the model factual(where costs evolve according to observed data) and the counterfactual scenario (where the US or China's trade or MP costs with all trading partners are fixed at 2016 levels). In all scenarios, knowledge stocks change endogenously. Changes are calculated as: (model factual − counterfactual) / model factual × 100%. Economies in both panels are ordered by the magnitude of the welfare effect in panel (a).

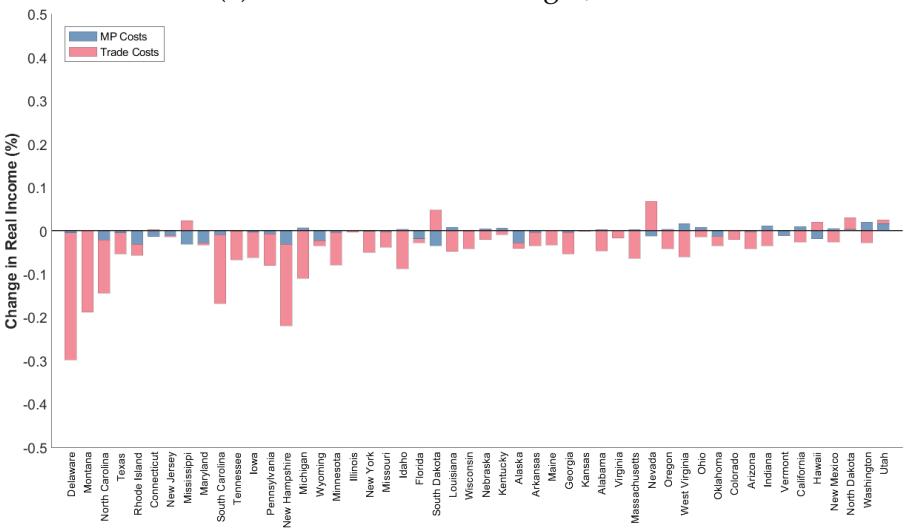
costs, that have a larger welfare effect.

The effects of these cost changes vary substantially across US states, with China-linked cost changes proving largely detrimental. Figure 8 reports the state-level counterparts for US- and China-linked costs. Note that for a given US state, holding all US-linked bilateral costs fixed is equivalent to holding all of its international costs fixed. Accordingly, panel (a) reports the contribution of US-linked costs, indicating the overall welfare effects are highly heterogeneous across states, and driven by both changes in MP costs and trade costs . Panel (b) singles out the contribution of China-linked costs. Here, however, trade cost changes play the dominant role. Moreover, the combined effect of changes in trade and MP costs with China is negative for most US states, suggesting that the shifts in bilateral costs with China over this period were welfare-reducing on net.

It is worth noting that the two counterfactual exercises can yield opposite welfare implications for the same country. For example, Australia experiences a welfare loss from changes in US-linked costs but a welfare gain from changes in China-linked costs. These heterogeneous responses reflect the fact that individual economies are differentially exposed to the US and China through distinct trade and MP channels. This observation naturally raises the question of whether countries that gain from proximity to one major economy tend to lose from proximity to the other. We take up this question in the next section.



(a) Gains from Cost Changes, 2016-2019



(b) Gains from Changes in Bilateral Costs with China, 2016-2019

Figure 8: Gains from Changes in Trade and MP Costs, by US State

Notes: This figure shows the welfare impact of the changes in US- and China-linked costs on each US state. Each bar reports the percentage change in real income between the model factual(where costs evolve according to observed data) and the counterfactual scenario (where the US or China's trade or MP costs with all trading partners are fixed at 2016 levels). In all scenarios, knowledge stocks change endogenously. Changes are calculated as: (model factual − counterfactual) / model factual × 100%. States in both panels are ordered by the magnitude of the welfare effect in panel (a).

6 Trade and MP Integration Patterns

To systematically examine whether proximity to one major economy comes at the expense of proximity to the other, we classify countries based on how their bilateral costs with the US and China have evolved over time. For each period (2000-2005, 2005-2016, 2016-2019), we classify countries into one of three integration groups—China-integrating, US-integrating, or non-integrating—based on two complementary approaches. We first examine changes in MP costs to determine group affiliation, and then apply the same classification framework using changes in trade costs. Our classification follows Bonadio et al. (2025), though we refer to these groups as "integrating" rather than "blocs" to better reflect the directional nature of the classification.

Specifically, for each t, we define the average MP costs between two countries $\overline{\gamma}_{ij,t}$ as the weighted average of the two bilateral MP costs, weighted by the respective MP flows,

$$\overline{\gamma}_{ij,t} = \frac{X^{MP}_{ij,t}}{X^{MP}_{ij,t} + X^{MP}_{ji,t}} \gamma_{ij,t} + \frac{X^{MP}_{ji,t}}{X^{MP}_{ij,t} + X^{MP}_{ji,t}} \gamma_{ji,t}.$$

Then, for each of the three periods, we compare how a country's MP costs with China and with the US change. Countries experiencing increased MP costs with China but decreased MP costs with the US are classified as US MP-integrating. Conversely, those with increased MP costs with the US but decreased MP costs with China are classified as China MP-integrating. Countries exhibiting neither pattern are classified as non-integrating.

$$i \in \begin{cases} \text{US MP-integrating} & \text{if } \Delta\overline{\gamma}_{i,CHN} \geq 0 \text{ and } \Delta\overline{\gamma}_{i,USA} < 0 \\ \text{China MP-integrating} & \text{if } \Delta\overline{\gamma}_{i,CHN} < 0 \text{ and } \Delta\overline{\gamma}_{i,USA} \geq 0 \\ \text{MP Non-integrating} & \text{otherwise.} \end{cases}$$

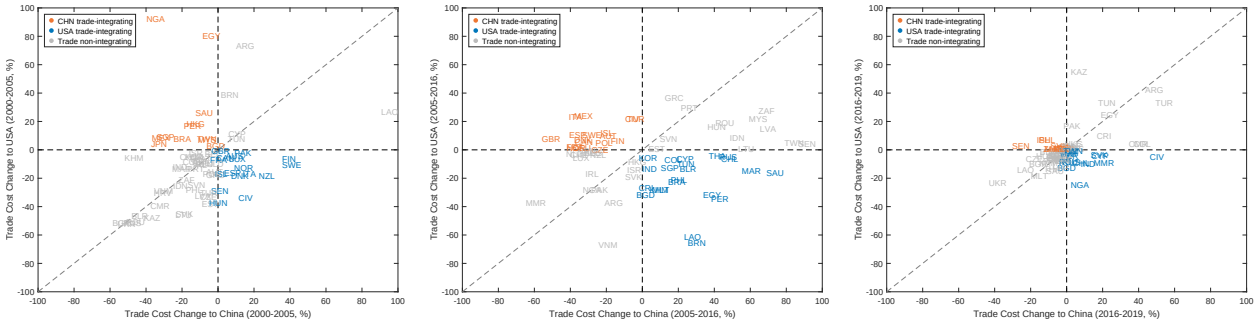
We apply this same classification framework to trade costs. We define the average trade costs between two economies as

$$\overline{\tau}_{ij,t} = \frac{X^{T}_{ij,t}}{X^{T}_{ij,t} + X^{T}_{ji,t}} \tau_{ij,t} + \frac{X^{T}_{ji,t}}{X^{T}_{ij,t} + X^{T}_{ji,t}} \tau_{ji,t},$$

where X^{T}_{ij} is i's import from j. We then classify economies into one of the three categories: US trade-integrating, China trade-integrating or trade non-integrating:

$$i \in \begin{cases} \text{US trade-integrating} & \text{if } \Delta\overline{\tau}_{i,CHN} \geq 0 \text{ and } \Delta\overline{\tau}_{i,USA} < 0 \\ \text{China trade-integrating} & \text{if } \Delta\overline{\tau}_{i,CHN} < 0 \text{ and } \Delta\overline{\tau}_{i,USA} \geq 0 \\ \text{Trade non-integrating} & \text{otherwise.} \end{cases}$$

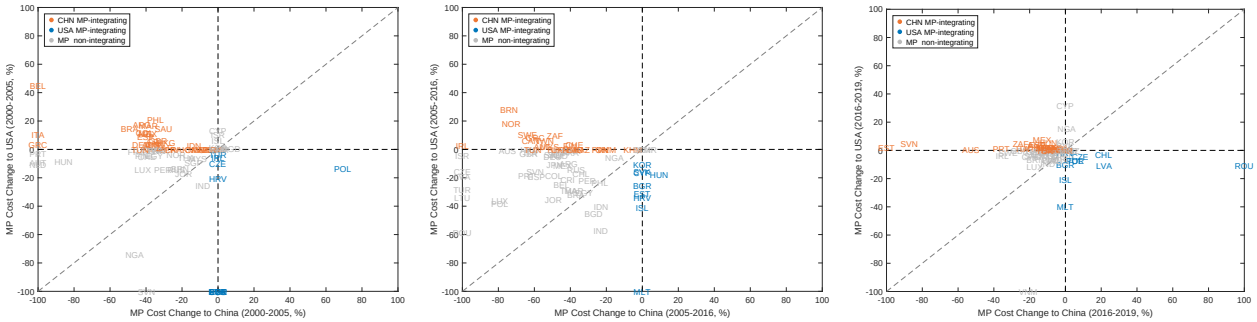
While trade and MP integration are often assumed to move in tandem, our analysis reveals a



(a) 2000-2005 (b) 2005-2016 (c) 2016-2019

Figure 9: Changes in Trade Costs to China or US

Notes: This figure shows the changes in trade costs between economies in different trade integration groups and China or the US in 3 different time periods. Trade costs between two economies are defined as the weighted average of two-way trade costs, weighed by the trade flow.



(a) 2000-2005 (b) 2005-2016 (c) 2016-2019

Figure 10: Changes in MP Costs to China or US

Notes: This figure shows the changes in MP costs between economies in different MP integration groups and China or the US in 3 different time periods. MP costs between two economies are defined as the weighted average of two-way MP costs, weighed by the MP flow.

far more complex landscape. Figures 9 and 10 show changes in each economy's trade and MP costs with China and the US. We see that economies do not necessarily have the same integration pattern in MP and in trade. Few economies are are non-integrating in both dimensions. More often, an economy may be China-integrating in MP but US-integrating or non-integrating in trade, or vice versa.

6.1 MP and Trade Integration Groups

Table 5 provides a detailed cross-tabulation of economies' MP and trade integration groups. The data shows that countries take different positions in multinational production and trade networks.

The first pattern is that very few economies are non-integrating in both dimensions simultaneously. Table 5 shows that in all three periods, the number of such economies is fewer than 19, or 25% of the total in our sample. The rarity of dual non-integration underscores how economic

interdependence has become a fundamental characteristic of the modern international system, with economies actively integrating through either trade or production networks to maintain competitiveness and growth.

(a) 2000-2005

MP/Trade	CHN	USA	Non	Total
CHN	6	9	18	33
USA	0	1	11	12
Non	8	9	14	31
Total	14	19	43	76

(b) 2005-2016

MP/Trade	CHN	USA	Non	Total
CHN	7	7	9	23
USA	1	4	5	10
Non	10	14	19	43
Total	18	25	33	76

(c) 2016-2019

MP/Trade	CHN	USA	Non	Total
CHN	7	6	20	33
USA	1	3	9	13
Non	5	9	16	30
Total	13	18	45	76

Table 5: MP and Trade Integration Groups

Notes: Cross-tabulation of economies' MP integration group (rows) versus trade integration group (columns) for each time period. CHN = China-integrating, USA = US-integrating, Non = Non-integrating. Diagonal entries indicate consistent classification across both dimensions; off-diagonal entries indicate divergent patterns.

Second, there exist significant inconsistencies between MP and trade integration patterns. Economies do not necessarily follow the same integration pattern across both dimensions. We can see it by looking at off diagonal entries in Table 5. This divergence takes various forms: an economy may be China-integrating in MP while being US-integrating or non-integrating in trade, or vice versa. The lack of strong correlation between MP and trade integration suggests that these two forms of economic linkage may serve distinct functions and are shaped by different factors, including technological capabilities, labor costs, market access, and geopolitical considerations.

In tables E.1 to E.3, we list the detailed integration group for each economy. Of particular interest

are cases where economies are China-integrating in MP while simultaneously US-integrating in trade. In 2000–2005, for example, while economies like Brazil, Mexico, and Japan are China-integrating in both dimensions, many others exhibit divergent patterns. Australia, Canada, Spain, and the United Kingdom, among others, were China MP-integrating and US trade-integrating at the same time. In 2005–2016, economies in the China MP-integrating / US trade-integrating group are mainly in East Europe and Asia, including Belarus, Brunei, Kazakhstan, Cambodia, Laos, and Ukraine. In 2016–2019, notably Bangladesh, Indonesia, Italy, and Myanmar fall into this group. Iceland is the only economy that is US MP-integrating and China trade-integrating in both 2005–2016 and 2016–2019.

To make sense of the China MP-integrating / US trade-integrating pattern—particularly evident among Asian and East European economies that serve as manufacturing hubs while maintaining strong trade relationships with the United States—a plausible explanation is that economies may find China's manufacturing and production capabilities essential for their industrial competitiveness, leading them to integrate into China-centered production networks. Simultaneously, these same economies may depend on US markets for exports or benefit from US-led trade agreements. This dual pattern allows economies to leverage China's production advantages while maintaining access to US consumer markets.

The observed patterns raise a provocative question about whether MP and trade integration function as substitutes rather than complements. The evidence of mismatched integration groups suggests that economies may indeed view these two forms of economic linkage as alternative pathways to achieving economic or political objectives. When economies face constraints or political pressures that limit their options in one dimension, they may compensate by deepening integration in the other. What is more, through the lens of our model where MP and trade integration entail different levels of knowledge spillovers, they may have very different long-run economic implications, which provides another rationale for economies to follow divergent integration paths.

6.2 Are Integration Patterns Persistent?

In the previous section, we established that the same economy may not exhibit consistent MP and trade integration patterns within the same period. In this section, we examine how economies transition across periods. How persistent is MP integration group membership? How about trade integration?

We start by summarizing the number of economies whose MP or trade integration group remains unchanged in Table 6. Two features stand out. First, most economies' integration patterns are fluid, shifting from one period to the next. This holds for both MP and trade classifications, with fewer than half of economies maintaining the same group from the preceding period. Second, MP integration groups exhibit slightly more stability than their trade counterparts, consistent with production networks being less easily reconfigured than trading relationships.

We next investigate whether persistence differs across groups. Table 7 provides a finer decom-

(a) MP Integration Group Transitions

Transition	Total Countries	Stable Countries	Stability Rate
2000-2005 to 2005-2016	76	35	0.461
2005-2016 to 2016-2019	76	33	0.434

(b) Trade Integration Group Transitions

Transition	Total Countries	Stable Countries	Stability Rate
2000-2005 to 2005-2016	76	23	0.303
2005-2016 to 2016-2019	76	32	0.421

Table 6: Transition Matrices for Integration Group Persistence Persistence

Notes: Transition matrices showing persistence and movement of economies between integration groups across consecutive time periods. Stable refers to economies maintaining the same integration group.

position. MP and trade integration transitions display distinct patterns. For MP, direct transitions between opposing integrating groups are rare. Most changes occur when economies move between the non-integrating group and either the China- or US-integrating group. China MP-integrating economies never transition directly to the US-integrating group, while transitions from US- to China-integrating are possible but infrequent. In contrast, for trade, direct transitions to the opposing group are much more common, especially in the earlier period. During the transition from 2000–2005 to 2005–2016, more than half of US trade-integrating economies shifted directly to the China-integrating group, while 42.9% shifted in the opposite direction. In the later period, direct transitions to the opposing group became uncommon, and economies were more likely to move to the non-integrating group.

Finally, the non-integrating category plays a notably different role across the two classifications. In MP, non-integrating economies maintain relatively stable positions (37.2 - 54.8% retention), with roughly balanced probabilities of joining either integrating group. In trade, the non-integrating category grows in importance over time: 55.6% of China trade-integrating and 60.0% of US trade-integrating economies transition to non-integrating in the later period. One interpretation is that this shift reflects increased trade diversification, particularly as US-China trade tensions intensified after 2016, though other factors, such as the maturation of regional trade agreements, may also contribute.

6.3 Alternative Integration

Economies do not always belong to the same integration group in both trade and MP. Some may be China- or US-integrating in trade while remaining non-integrating in MP. Others may exhibit opposing patterns, such as US-integrating in trade while China-integrating in MP. Do economies

(a) MP: 2000–2005 to 2005–2016

From/To	CHN	USA	Non
CHN	0.394	0.000	0.606
USA	0.083	0.417	0.500
Non	0.290	0.161	0.548

(b) MP: 2005–2016 to 2016–2019

From/To	CHN	USA	Non
CHN	0.522	0.000	0.478
USA	0.200	0.500	0.300
Non	0.442	0.186	0.372

(c) Trade: 2000–2005 to 2005–2016

From/To	CHN	USA	Non
CHN	0.143	0.429	0.429
USA	0.579	0.053	0.368
Non	0.116	0.419	0.465

(d) Trade: 2005–2016 to 2016–2019

From/To	CHN	USA	Non
CHN	0.222	0.222	0.556
USA	0.080	0.320	0.600
Non	0.212	0.182	0.606

Table 7: Detailed Transition Matrices for Integration Groups

Notes: Detailed transition matrices showing economy counts moving between integration groups across consecutive periods. Numbers are proportions of economies.

benefit from aligning their trade and MP integration patterns?

To answer this question, we perform counterfactual exercises in which we reassign economies to different integration groups and compare their welfare to the baseline. We follow Bonadio et al. (2025) and assume that when an economy joins an integration group, its bilateral trade and MP costs with all other economies change in a manner similar to those observed among economies already in that group.

6.3.1 *Counterfactual Design*

We describe how we set counterfactual tariffs as follows; counterfactual iceberg trade costs and MP costs are constructed analogously.[9] Denote the tariff component in trade costs as $\delta_{ij,t} = 1 + t_{ij,t}$ and the change rate from the beginning of the period ($t0$) to the end ($t1$) as $\dot{\delta}_{ij} = \frac{\delta_{ij,t1} - \delta_{ij,t0}}{\delta_{ij,t0}}$. For simplicity, time subscripts are suppressed from the following formulas. We first calculate the average import and export tariff changes between each economy m and a given integration group b. The average import cost change from economy m to group b is:

[9]We use MP flows (X^{MP}_{mn}) instead of trade flows (X_{mn}) to construct the weights for the average change rate in MP costs.

$$\dot{\delta}_{bm} = \begin{cases} \dfrac{1}{N_b - \mathbf{1}_{\{m \in b\}}} \displaystyle\sum_{\substack{n \in b \cap C \\ n \neq m}} \dot{\delta}_{nm}, & \text{if } m \in C, \\ \dfrac{1}{N_b - \mathbf{1}_{\{m \in b\}}} \displaystyle\sum_{\substack{n \in b \cap C \\ n \neq \text{USA}}} \dot{\delta}_{nm}, & \text{if } m \in S, \end{cases}$$

where

$$\dot{\delta}_{\text{USA},m} = \frac{1}{|S|} \sum_{s \in S} \dot{\delta}_{sm}.$$

And the average export cost change from group b to economy m:

$$\dot{\delta}_{mb} = \begin{cases} \dfrac{1}{N_b - \mathbf{1}_{\{m \in b\}}} \displaystyle\sum_{\substack{n \in b \cap C \\ n \neq m}} \dot{\delta}_{mn}, & \text{if } m \in C, \\ \dfrac{1}{N_b - \mathbf{1}_{\{m \in b\}}} \displaystyle\sum_{\substack{n \in b \cap C \\ n \neq \text{USA}}} \dot{\delta}_{mn}, & \text{if } m \in S, \end{cases}$$

where

$$\dot{\delta}_{m,\text{USA}} = \frac{1}{|S|} \sum_{s \in S} \dot{\delta}_{ms}.$$

Thus, we obtain the cost changes for any given economy-group pair, reflecting the cost structure applied by a group toward an economy and vice versa. When assuming an economy joins a particular integration group and adopts its cost structure, we do not simply apply the group's average cost changes, as this would ignore the economy's own fixed effect in cost changes during the period. Instead, we use trade volume as weights to calculate an adjustment factor based on the difference between the economy's actual average cost changes and the hypothetical average under the group's cost structure. This ensures that the trade-weighted average cost change for the economy is preserved under the counterfactual. The import and export adjustment factors when economy n joins group b are:

$$\dot{\delta}^{b,\text{adj}}_{n,im} = \sum_{m \neq n} \frac{X_{nm}}{\sum_{m \neq n} X_{nm}} \dot{\delta}_{nm} - \sum_{m \neq n} \frac{X_{nm}}{\sum_{m \neq n} X_{nm}} \dot{\delta}_{bm}$$

$$\dot{\delta}^{b,\text{adj}}_{n,ex} = \sum_{m \neq n} \frac{X_{mn}}{\sum_{m \neq n} X_{mn}} \dot{\delta}_{mn} - \sum_{m \neq n} \frac{X_{mn}}{\sum_{m \neq n} X_{mn}} \dot{\delta}_{mb}$$

where X_{nm} represents the trade volume that economy n purchases from economy m at the beginning of the period. The counterfactual trade cost change rate when placing economy n into group b is:

$$\dot{\delta}^{\text{CF},b}_{n,m} = \dot{\delta}_{b,m} + \dot{\delta}^{b,\text{adj}}_{n,im}$$

$$\dot{\delta}^{\text{CF},b}_{m,n} = \dot{\delta}_{m,b} + \dot{\delta}^{b,\text{adj}}_{n,ex}$$

This approach ensures that after economy n joins group b, the trade-volume-weighted average change rate of its counterfactual import costs matches the actual scenario, and the ordering of cost change rates applied by n to all partners mirrors that of the average group b member. The same properties hold symmetrically for export costs.

The counterfactual examines the impact of adjusting an economy's trade or MP cost structure to match that of integration group b, while keeping its average cost change constant. We adjust one economy at a time, maintaining the trade and MP cost change rates for all other economies at their actual observed values.

Our counterfactual focuses on the most recent period in our sample (2016-2019) and beyond. We assume that economies did not follow their actual integration patterns in 2016 but instead followed the counterfactually assigned patterns. After calculating the counterfactual costs in 2019, we compare the welfare difference between the counterfactual and actual cost scenarios to measure the static gains:

$$\text{Static Gains} = \log\left(\frac{C^{\text{CF}}_{i,2019}}{C^{\text{Benchmark}}_{i,2019}}\right),$$

where C_i is aggregate consumption per worker in economy i. We then run the economy forward from 2019 with cost parameters fixed at their 2019 values, allowing knowledge stocks to evolve endogenously. The benchmark uses actual trade and MP costs in 2019, while the counterfactual uses counterfactual costs. We define the welfare gain from switching integration groups as the annual consumption equivalent variation:

$$\text{Welfare} = (1-\beta)\sum_j \beta^j \log\left(\frac{C^{\text{CF}}_{i,2019+j}}{C^{\text{Benchmark}}_{i,2019+j}}\right).$$

6.3.2 *Welfare Effects of Integration Group Realignment*

We consider economies whose trade and MP integration groups differ in 2016–2019 and ask what would happen if they achieved consistent integration—aligning both dimensions with the same group. We examine six configurations based on initial MP and trade integration status: split alignments between China and US (in both directions), and cases where one dimension is integrating and the other is non-integrating (four combinations). Full country-level results for each configuration are reported in Appendix Tables F.1–F.5.

Figure 11 summarizes the results. Each point represents an economy; the horizontal axis measures the static gain and the vertical axis measures the welfare effect from switching to full alignment with either China (left panel) or the US (right panel). Three patterns emerge.

First, the majority of economies lie in the third quadrant, exhibiting both negative static gains and negative welfare effects. For most economies, the current pattern of misaligned integration appears economically rational: forcing consistent integration would reduce welfare in both the short and long run.

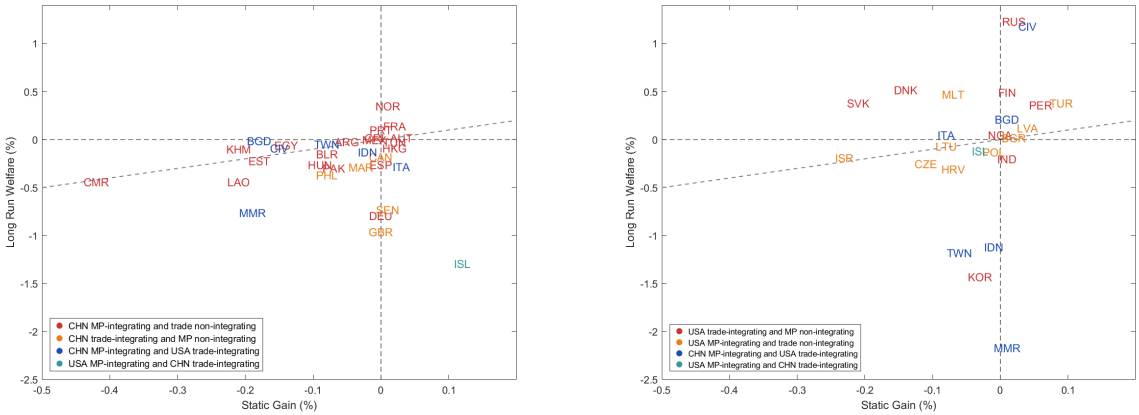


Figure 11: Welfare Effects of Integration Group Realignment

Notes: This figure displays static and welfare gains from integration group realignment. The left panel presents results for economies switching to full China-integrating status. For economies initially China MP-integrating but not China trade-integrating, this entails switching to the China trade-integrating group. For economies initially China trade-integrating but not China MP-integrating, this entails switching to the China MP-integrating group. The right panel presents analogous results for economies switching to full US-integrating status. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

Second, for economies switching to full China integration, welfare effects are generally smaller than static effects, indicating negative growth effects. This is consistent with the model's mechanism: realigning production and trade relationships alters the channels through which technology diffuses across borders, and for this group the disruption to knowledge accumulation more than offsets any static efficiency gains.

Third, a handful of economies would benefit from consistent integration. On the China-integrating side, examples include Austria, France, and Norway. On the US-integrating side, Latvia, Turkey, and Peru would experience positive welfare gains from full alignment. For these economies, the current misalignment may reflect considerations beyond static and dynamic economic efficiency.

The results also reveal a trade-off between static efficiency and dynamic growth. Iceland and Italy achieve positive static gains from adding China trade integration, yet overall welfare turns negative once growth effects are accounted for. This underscores that static cost savings from integration group consolidation can be offset, and even reversed, by impaired technology diffusion in the long run.

6.3.3 *Tariff Changes*

We also examine a variant of this counterfactual in which only the tariff component of trade costs is adjusted, leaving non-tariff components unchanged (Appendix Tables F.6–F.8). The qualitative patterns are similar, but both static gains and welfare effects are an order of magnitude smaller. There are two reasons. First, the primary drivers of recent trade cost changes are non-tariff

components such as transportation costs, non-tariff barriers, and geopolitical frictions (Fan et al., 2025). Second, the tariff escalation between the U.S. and China was still in its early stages during our sample period; more substantial increases occurred well after 2019 and are therefore not captured in our baseline cost estimates. Extending the analysis is an important direction for future work.

We perform additional exercises for non-tariff changes in the Appendix G to examine whether tariff or non-tariff components dominate during the realignment of trade integration groups.

	Welfare(%)			Static Gains(%)		
	Median	p25	p75	Median	p25	p75
Both Change	-0.095	-0.260	0.010	-0.040	-0.145	0.015
Tariff Change	-0.010	-0.050	0.015	0.000	-0.010	0.010
Non-tariff Change	-0.060	-0.205	0.055	-0.015	-0.125	0.025

Table 8: Welfare effect of Trade Integrating Realignment

Notes: This table reports the cross-country distribution of welfare effects when trade integrating realignment. Columns report the median, 25th percentile, and 75th percentile across economies."Both Change" reports welfare effects which tariff component and non-tariff components trade costs are adjusted. "Tariff Change" reports welfare effects which only the tariff component of trade costs is adjusted, leaving non-tariff components unchanged. "Non-tariff Change" reports welfare effects which only the non-tariff component of trade costs is adjusted, leaving tariff components unchanged. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

Table 8 report the welfare effect when trade integration realigns. The static gains are far smaller than the total welfare effects, implying that absent knowledge diffusion, welfare estimation would be severely biased. The medians are negative across all three scenarios, indicating that most countries are economically rational when selecting their trade integration group. Non-tariff component accounts for the dominant share of welfare variation, while the effect from only adjusting the tariff component is an order of magnitude smaller.

6.3.4 Subnational Reorientation: California

The preceding analysis examines integration group realignment at the country level. We now ask whether a similar framework can shed light on subnational reorientation. Specifically, we consider a counterfactual in which California, the largest US state economy and one with deep trade and investment ties to both China and the rest of the United States, shifts from the US-integrating group to the China-integrating group.[10]

[10]California is the first state to legally challenge the president's use of emergency powers to impose import duties. See https://www.gov.ca.gov/2025/04/16/governor-newsom-files-lawsuit-to-end-president-trumps-tariffs/

We perform a counterfactual exercise where California's costs with all other US states remain unchanged, but its costs with all other countries adjust as if it had joined the China-integrating group. This isolates the effect of international reorientation while preserving domestic integration. Table 9 column (1) shows that the overall welfare gain is 0.17%, the majority of which comes from trade integration with China rather than MP integration.

	Welfare (1)	Static Gain (2)
MP	0.04%	-0.02%
Trade	0.14%	0.04%
Trade and MP	0.17%	0.01%

Table 9: Welfare Effects of California Joining the China-Integrating Group

Notes: This table presents welfare effects of California joining the China-integrating group, holding California's costs with all other US states unchanged. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

7 Concluding Remarks

This paper studies the welfare and growth consequences of realignment in global production and trade. We develop a quantitative framework that embeds multinational production and technology diffusion into a multi-country, semi-endogenous growth model with subnational geography. We also document that while trade between the US and China fragmented after 2016, China's outward multinational production continued to expand globally, including into the US.

Counterfactual exercises reveal that MP cost changes account for the dominant share of welfare variation across countries, and that endogenous knowledge accumulation substantially amplifies the effects of cost changes relative to a purely static analysis. At the subnational level, we find that US states exhibit significant heterogeneity in welfare outcomes, driven not only by differences in cost changes but also by differences in underlying economic structure. We further show that countries' MP and trade integration patterns frequently diverge. An economy may integrate with China in multinational production while integrating with the US in trade—and that forcing alignment across these dimensions would predominantly reduce welfare, as short-run efficiency gains from consolidation are offset by impaired knowledge diffusion in the long run.

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A Theory

In this section we present additional derivations for Section 2.

A.1 Cumulated Knowledge Stock

We first derive the augmented arrival rate. Under Assumption of original compeonent $\boldsymbol{Z}$ and taking the distribution of $\boldsymbol{Q}'_p$ $\boldsymbol{Q}'_s$ given, the arrival rate of ideas of quality $\boldsymbol{Q}$ better than $\boldsymbol{q}$ in at least one dimension is

$$\tilde{A}_{h,t}(\boldsymbol{q}) = \int_0^\infty \int_0^\infty A_{h,t}\left(\frac{\boldsymbol{q}}{x^{\rho_1} y^{\rho_2}}\right) dG_{h,t}(x) dH_{h,t}(y)$$

$$= \int_0^\infty \int_0^\infty \alpha_{h,t}\left(\sum_{n=1}^{N}\left(\frac{q_n}{x^{\rho_1} y^{\rho_2}}\right)^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} dG_{h,t}(x) dH_{h,t}(y)$$

$$= \alpha_{h,t}\left(\sum_{n=1}^{N} q_n^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} \int_0^\infty \int_0^\infty (x^{\rho_1} y^{\rho_2})^\theta dG_{h,t}(x) dH_{h,t}(y).$$

We define $\Lambda_{h,t} = \int_0^\infty \int_0^\infty (x^{\rho_1} y^{\rho_2})^\theta dG_{h,t}(x) dH_{h,t}(y)$ as the learning effect.

Then we derive the law of motion of knowledge stock. Denote the number of ideas with $\boldsymbol{Q}$ better than $\boldsymbol{q}$ for producing good v in at least one dimension as $M_{h,t}$. This number $M_{h,t}$ is a random variable and follows a Poisson distribution with parameter

$$\int_{-\infty}^{t} \tilde{A}_{h,\tau}(\boldsymbol{q}) d\tau.$$

Observe that the event "state-of-the-art efficiencies to produce good v, $\boldsymbol{Q}_{h,t}^{(1)}(v) \leq \boldsymbol{q}$" is equivalent to the event "number of ideas better than q in at least one dimension $M_{h,t} = 0$". Hence,

$$\Pr[\boldsymbol{Q}_{h,t}^{(1)}(v) \leq \boldsymbol{q}] = \Pr[M_{h,t} = 0] = \exp\left(-\int_{-\infty}^{t} \tilde{A}_{h,\tau}(\boldsymbol{q}) d\tau\right).$$

It follows immediately that

$$\Pr[\boldsymbol{Q}_{h,t}^{(1)}(v) \leq \boldsymbol{q}] = \exp\left(-\int_{-\infty}^{t} \tilde{A}_{h,\tau}(\boldsymbol{q}) d\tau\right)$$

$$= \exp\left(-\int_{-\infty}^{t} \alpha_{h,\tau}\Lambda_{h,\tau}\left(\sum_{n=1}^{N} q_n^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} d\tau\right)$$

$$= \exp\left(-\int_{-\infty}^{t} \alpha_{h,\tau}\Lambda_{h,\tau} d\tau \left(\sum_{n=1}^{N} q_n^{-\frac{\theta}{1-\eta}}\right)^{1-\eta}\right)$$

This implies that the frontier of knowledge retains its shape, and the scale parameter takes the form $\lambda_{h,t} = \int_{-\infty}^{t} \alpha_{h,\tau} \Lambda_{h,\tau} d\tau$.

It follows that

$$\dot{\lambda}_{h,t} = \alpha_{h,t} \int_0^\infty \int_0^\infty \left(x^{\rho_1} y^{\rho_2}\right)^\theta dG_{h,t}(x)\, dH_{h,t}(y),$$

where $G_{h,t}(x)$ and $H_{h,t}(y)$ are technology frontiers of producers and sellers, respectively.

A.2 Law of Motion for Knowledge Stock

We first derive the distribution of efficiencies of actively producing plants located in country h at time t, $G_{h,t}(\cdot)$, and the distribution of efficiencies of producers that sell goods to country h at time t, $H_{h,t}(\cdot)$, then we derive the evolution of knowledge stock. In this proof, we will omit time subscript t when there is no confusion.

Let $K_{ijh}(q)$ be the fraction of goods for which the lowest cost provider to i is a plant in j whose headquarter is in h, and the productivity of this plant is weakly less than q.

If insights from producer are randomly and uniformly drawn from the distribution of productivity among all prodcuing plants in h, then

$$G_h(q) = \frac{\sum_{n=1}^{N} K_{hhn}(q)}{\sum_{n=1}^{N} K_{hhn}(\infty)}.$$

Similarly, we assume insights from seller are randomly and uniformly drawn from all producers that sell goods to country h when a new idea arrives, then

$$H_h(q) = \sum_{j=1}^{N} \sum_{n=1}^{N} K_{hjn}(q).$$

By definition,

$$K_{ijh}(q) = \Pr\left(q_{jh} \leq q \& q_{kn} \leq \frac{\kappa_{ikn}}{\kappa_{ijh}} q_{jh}\ \forall k, n\right)$$

$$= \int_0^q \frac{\partial F_h\left(\frac{\kappa_{i1h}}{\kappa_{ijh}} x, \ldots, \frac{\kappa_{iNh}}{\kappa_{ijh}} x\right)}{\partial q_{jh}} \left(\prod_{n \neq h} F_n\left(\frac{\kappa_{i1n}}{\kappa_{ijh}} x, \ldots, \frac{\kappa_{iNn}}{\kappa_{ijh}} x\right)\right) dx,$$

where $\kappa_{ijh} = \gamma_{jh} w_j \tau_{ij}$.

From Appendix A.1, we know that $(q_{1h}, \ldots, q_{Nh})$ follows a multivariate Fréchet distribution with location parameter λ_h, $F_h(\cdot)$, so

$$F_n\left(\frac{\kappa_{i1n}}{\kappa_{ijh}} x, \ldots, \frac{\kappa_{iNn}}{\kappa_{ijh}} x\right) = \exp\left(-\lambda_n \left(\sum_{k=1}^{N} \left(\frac{\kappa_{ikn}}{\kappa_{ijh}}\right)^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} x^{-\theta}\right).$$

And

$$\frac{\partial F_h(q)}{\partial q_{jh}} = \exp\left(-\lambda_h \left(\sum_{k=1}^{N} q_{kh}^{-\frac{\theta}{1-\eta}}\right)^{1-\eta}\right) \theta\lambda_h \left(\sum_{k=1}^{N} q_{kh}^{-\frac{\theta}{1-\eta}}\right)^{-\eta} q_{jh}^{-\frac{\theta}{1-\eta}-1},$$

so

$$K_{ijh}(q) = \int_0^q \theta\lambda_h \left(\sum_{k=1}^{N} \left(\frac{\kappa_{ikh}}{\kappa_{ijh}} x\right)^{-\frac{\theta}{1-\eta}}\right)^{-\eta} x^{-\frac{\theta}{1-\eta}-1} \prod_{n=1}^{N} \exp\left(-\lambda_n \left(\sum_{k=1}^{N} \left(\frac{\kappa_{ikn}}{\kappa_{ijh}}\right)^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} x^{-\theta}\right) dx$$

$$= \int_0^q \theta\lambda_h \left(\sum_{k=1}^{N} \left(\frac{\kappa_{ikh}}{\kappa_{ijh}}\right)^{-\frac{\theta}{1-\eta}}\right)^{-\eta} \prod_{n=1}^{N} \exp\left(-\lambda_n \left(\sum_{k=1}^{N} \left(\frac{\kappa_{ikn}}{\kappa_{ijh}}\right)^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} x^{-\theta}\right) x^{-\theta-1} dx$$

$$= \int_0^q (-\lambda_h) \left(\sum_k (\frac{\kappa_{ikh}}{\kappa_{ijh}})^{-\frac{\theta}{1-\eta}}\right)^{-\eta} \exp\left(\sum_{n=1}^{N} -\lambda_n \left(\sum_{k=1}^{N} (\frac{\kappa_{ikn}}{\kappa_{ijh}})^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} x^{-\theta}\right) dx^{-\theta}$$

$$= \frac{\lambda_h \left(\sum_k \left(\frac{\kappa_{ikh}}{\kappa_{ijh}}\right)^{-\frac{\theta}{1-\eta}}\right)^{-\eta}}{\sum_{n=1}^{N} \lambda_n \left(\sum_{k=1}^{N} \left(\frac{\kappa_{ikn}}{\kappa_{ijh}}\right)^{-\frac{\theta}{1-\eta}}\right)^{1-\eta}} \exp\left(\sum_{n=1}^{N} -\lambda_n \left(\sum_{k=1}^{N} \left(\frac{\kappa_{ikn}}{\kappa_{ijh}}\right)^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} q^{-\theta}\right)$$

$$= \kappa_{ijh}^{-\frac{\theta}{1-\eta}} \frac{\lambda_h \left(\sum_k (\kappa_{ikh})^{-\frac{\theta}{1-\eta}}\right)^{-\eta}}{\sum_{n=1}^{N} \lambda_n \left(\sum_{k=1}^{N} (\kappa_{ikn})^{-\frac{\theta}{1-\eta}}\right)^{1-\eta}} \exp\left(\sum_{n=1}^{N} -\lambda_n \left(\sum_{k=1}^{N} (\frac{\kappa_{ikn}}{\kappa_{ijh}})^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} q^{-\theta}\right)$$

$$= \pi_{ijh} \exp\left(\sum_{n=1}^{N} -\lambda_n \left(\sum_{k=1}^{N} (\frac{\kappa_{ikn}}{\kappa_{ijh}})^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} q^{-\theta}\right).$$

Note that

$$\sum_{n=1}^{N} -\lambda_n \left(\sum_{k=1}^{N} \left(\frac{\kappa_{ikn}}{\kappa_{ijh}}\right)^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} = \sum_{n=1}^{N} \lambda_n \left(\sum_{k=1}^{N} (\kappa_{ikn})^{-\frac{\theta}{1-\eta}}\right)^{1-\eta} \kappa_{ijh}^{\theta}$$

$$= \frac{\lambda_h}{\pi_{ijh}\psi_{ijh}^{-\eta}}$$

$$= \frac{\lambda_h}{\pi_{ih}^{E}\psi_{ijh}^{1-\eta}}.$$

So we have

$$K_{ijh}(q) = \pi_{ijh} \exp\left(-\frac{\lambda_h}{\pi^E_{ih}\psi^{1-\eta}_{ijh}} q^{-\theta}\right),$$

and

$$G_h(q) = \frac{\sum_{n=1}^N K_{hhn}(q)}{\sum_{n=1}^N K_{hhn}(\infty)} = \sum_{n=1}^N \frac{\pi_{hhn}}{\sum_{n=1}^N \pi_{hhn}} \exp\left(-\frac{\lambda_n}{\pi^E_{hn}\psi^{1-\eta}_{hhn}} q^{-\theta}\right),$$

$$H_h(q) = \sum_{j=1}^N \sum_{n=1}^N K_{hjn}(q) = \sum_{j=1}^N \sum_{n=1}^N \frac{\pi_{hjn}}{\sum_{j=1}^N \sum_{n=1}^N \pi_{hjn}} \exp\left(-\frac{\lambda_n}{\pi^E_{hn}\psi^{1-\eta}_{hjn}} q^{-\theta}\right).$$

Finally, substituting $G_h(q)$ and $H_h(q)$ back to

$$\Lambda_{h,t} = \int_0^\infty \int_0^\infty (x^{\rho_1} y^{\rho_2})^\theta dG_{h,t}(x) dH_{h,t}(y),$$

and recall that

$$\dot{\lambda}_{h,t} = \alpha_{h,t}\Lambda_{h,t},$$

We thus get the law of motion of knowledge stock:

$$\dot{\lambda}_{h,t} = \alpha_{h,t}\Gamma_{\rho_1,\rho_2}\left[\sum_n \frac{\pi_{hhn,t}}{\sum_n \pi_{hhn,t}}\left(\frac{\lambda_{n,t}}{\pi^E_{hn,t}\psi^{1-\eta}_{hnn,t}}\right)^{\rho_1}\right]\left[\sum_n\sum_j \frac{\pi_{hjn,t}}{\sum_n\sum_j \pi_{hjn,t}}\left(\frac{\lambda_{n,t}}{\pi^E_{hn,t}\psi^{1-\eta}_{hjn,t}}\right)^{\rho_2}\right].$$

B Calibration Algorithm

The calibration procedure recovers bilateral trade costs τ_{ij}, MP costs γ_{jh}, and knowledge stocks λ_i from the observed dataset $\mathcal{D}$ and the interstate MP cost parameter γ_{state}. The procedure involves three nested layers. At the outermost layer, γ_{state} is chosen to minimize the distance between model-implied and observed MP self-shares for U.S. states. Each evaluation of this objective requires a full model solve, which itself involves two nested fixed-point problems. The outer fixed point iterates over knowledge stocks λ_i: conditional on a guess for λ_i, an inner fixed point recovers bilateral MP and trade costs (γ_{jh}, τ_{ij}) from the bilateral share equations and the constraints on γ; the trilateral shares π_{ijh} implied by these costs are then used to update λ_i via equation (6), and the process repeats until convergence.

We present the algorithms bottom-up. We first describes the inner fixed point that recovers costs conditional on knowledge stocks. Then we describe the outer fixed point which recovers knowledge stocks. Finally, we describe the estimation of γ_{state}.

B.1 Inner Loop

The algorithm to solve the model given λ_i, t_{ij}, some constraints on γ, and an allocation of the economy $\{\pi^T_{ij}, X_i, w_i, Y_i\}$, together with part of the data for π^M_{jh}, is as follows:

1. Initiate the algorithm with a guess $\pi^{M(0)}_{jh}$. For the known entries of π^M_{jh}, $\pi^{M(0)}_{jh}$ should match the given π^M_{jh}.

2. Use λ_i, t_{ij}, π^T_{ij}, $\pi^{M(0)}_{jh}$, X_i, w_i, and Y_i to solve for $\gamma^{(0)}$:

   (a) Initiate the algorithm with a guess of $g^{(0)}$ and $h^{(0)}$, and make the diagonal elements of $g^{(0)}$ be λ, and make the diagonal elements of $h^{(0)}$ be $w^{-\theta}$, where the superscript (0) indicates that it is a guess.

   (b) Use $g^{(0)}$, $h^{(0)}$, X_i, π^T_{ij}, π^M_{jh}, Y_i, and t_{ij} to solve for $g^{(1)}$ and $h^{(1)}$ using the following equations, where the superscript (1) indicates an updated value for g and h.

$$h^{(1)}_{ij} = \left(\sum_{h=1}^{N} \frac{(g^{(0)}_{jh})^{1/(1-\eta)}}{\pi^T_{ij} \sum_{k=1}^{N} (h^{(0)}_{ik} g^{(0)}_{kh})^{1/(1-\eta)}} \frac{\left(\sum_{k=1}^{N} (h^{(0)}_{ik} g^{(0)}_{kh})^{1/(1-\eta)} \right)^{1-\eta}}{\sum_{n=1}^{N} \left(\sum_{k=1}^{N} (h^{(0)}_{ik} g^{(0)}_{kn})^{1/(1-\eta)} \right)^{1-\eta}} \right)^{1-\eta},$$

$$g^{(1)}_{jh} = \left(\sum_{i=1}^{N} \frac{X_i \frac{1}{1+t_{ij}} (h^{(0)}_{ij})^{1/(1-\eta)}}{\pi^M_{jh} Y_j \sum_{k=1}^{N} (h^{(0)}_{ik} g^{(0)}_{kh})^{1/(1-\eta)}} \frac{\left(\sum_{k=1}^{N} (h^{(0)}_{ik} g^{(0)}_{kh})^{1/(1-\eta)} \right)^{1-\eta}}{\sum_{n=1}^{N} \left(\sum_{k=1}^{N} (h^{(0)}_{ik} g^{(0)}_{kn})^{1/(1-\eta)} \right)^{1-\eta}} \right)^{1-\eta}.$$

   (c) Check if $g^{(0)} \simeq g^{(1)}$ and $h^{(0)} \simeq h^{(1)}$. If it is not, make the diagonal elements of $g^{(1)}$ be λ, and make the diagonal elements of $h^{(1)}$ be $w^{-\theta}$ while keeping all other elements unchanged. Use the resulting $g^{(1')}$ and $h^{(1')}$ as the initial guess and go back to step 1.

   (d) Use g and h solved above to solve for γ_{jh}, τ_{ij}, and π_{ijh} using the following equations:

$$\pi_{ijh} = \frac{(h_{ij} g_{ih})^{1/(1-\eta)}}{\sum_{k=1}^{N} (h_{ik} g_{kh})^{1/(1-\eta)}} \frac{\left(\sum_{k=1}^{N} (h_{ik} g_{kh})^{1/(1-\eta)} \right)^{1-\eta}}{\sum_{n=1}^{N} \left(\sum_{k=1}^{N} (h_{ik} g_{kn})^{1/(1-\eta)} \right)^{1-\eta}},$$

$$\gamma_{jh} = \left(\frac{\lambda_h}{g_{jh}} \right)^{1/\theta},$$

$$\tau_{ij} = \frac{h_{ij}^{-1/\theta}}{w_j}.$$

3. Keep the unconstrained part of $\gamma^{(0)}$ unchanged, and make minimal modifications to $\gamma^{(0)}$ to obtain $\gamma^{(1)}$ such that it satisfies the constraints on γ. The modification rule must satisfy: if $\gamma^{(0)}$ already meets the constraints, then $\gamma^{(1)} = \gamma^{(0)}$.

4. Use λ_i, t_{ij}, $\gamma^{(1)}$, X_i w_i, Y_i and h_{ij} solved from step 2 to solve for $\pi_{jh}^{M(1)}$ using the following equations.

$$\pi_{jh}^{M(1)} = \sum_{i=1}^{N} \frac{X_i \frac{1}{1+t_{ij}} (h_{ij} g_{jh}^{(1)})^{1/(1-\eta)}}{Y_j \sum_{k=1}^{N} (h_{ik} g_{kh}^{(1)})^{1/(1-\eta)}} \frac{\left(\sum_{k=1}^{N} (h_{ik} g_{kh}^{(1)})^{1/(1-\eta)}\right)^{1-\eta}}{\sum_{n=1}^{N} \left(\sum_{k=1}^{N} (h_{ik} g_{kn}^{(1)})^{1/(1-\eta)}\right)^{1-\eta}},$$

where

$$g_{jh}^{(1)} = (\gamma_{jh}^{(1)})^{-\theta} \lambda_h.$$

5. Check whether $\pi_{jh}^{M(0)} \simeq \pi_{jh}^{M(1)}$ and $\gamma^{(0)} = \gamma^{(1)}$. If not, modify $\pi_{jh}^{M(1)}$ by replacing its known entries with the corresponding values from the data (keeping other elements unchanged) and normalize each row so that the row sum equals 1. Then use the resulting $\pi_{jh}^{M(1')}$ as the new initial guess and go back to step 1.

B.2 Outer Loop

The algorithm to solve the model given t_{ij}, some constraints on γ, and an allocation of the economy $\{\pi_{ij}^T, X_i, w_i, \omega_i, Y_i\}$, together with part of the data for π_{jh}^M, is as follows:

1. Initiate the algorithm with a guess $\lambda^{(0)}$, where the superscript (0) indicates that it is a guess.

2. Use $\lambda^{(0)}$, t_{ij}, constraints on γ, and the given allocation of the economy $\{\pi_{ij}^T, X_i, w_i, Y_i\}$ together with part of the data for π_{jh}^M to solve for π_{ijk} in the *inner loop*.

3. Use π_{ijk} from step 2 and ω from PWT to solve for $\lambda^{(1)}$ using the following equation, where the superscript (1) indicates an updated value for λ:

$$\lambda_i^{(1)} = B^{-\theta} (\omega_i)^{\theta} \pi_{iii} \psi_{iii}^{-\eta}.$$

4. Check if $\lambda^{(1)} \simeq \lambda^{(0)}$. If not, take $\lambda^{(1)}$ as the new initial guess and go back to step 1.

B.3 Estimation of γ_{state}

Given the parameter γ_{state} and the dataset $\mathcal{D}$, all remaining parameters and MP shares can be recovered by solving the system of equations defined by (1)–(6). The following details the procedure to pin down γ_{state} .

The interstate MP cost γ_{state} governs the effective productivity of a state's knowledge used in other states. Varying γ_{state} significantly alters the model's predictions for **(i)** the share of production using its own knowledge (the "MP self-share"); **(ii)** the share of production using knowledge from other states. To find γ_{state}, we introduce additional data not contained in $\mathcal{D}$: an empirical target for the MP self-share of U.S. states. We then determine the optimal γ^*_{state} by minimizing a weighted distance between the model-implied and the data-based MP self-shares, ensuring the model's internal MP cost aligns with the external empirical benchmark.

We observe, for each U.S. state s, the share of total establishments accounted by three firm types from (Kamal et al., 2022) [11]: U.S. multinational enterprises (MNEs), denoted $\pi^{\text{MNE}}_{s,\text{U.S.}}$; U.S. non-MNEs, denoted $\pi^{\text{non-MNE}}_{s,\text{U.S.}}$; and foreign MNEs, denoted $\pi^{\text{MNE}}_{s,\text{foreign}}$.

We assume that for each state, the production share of each type of firm equals its corresponding establishment share. Non-MNE firms are assumed to have their headquarters in the same state where production occurs, whereas the headquarters of U.S. MNEs may be located in a different state from production, and a state's own MNEs production share of aggregate U.S. MNEs is equal to that state's real GDP share of aggregate U.S. real GDP, $\pi^{\text{GDP}}_{s,\text{U.S.}}$. Formally, the observed self MP share is given by:

$$\pi^M_{ss} = \pi^{M,\text{non-mne}}_{ss} + \pi^{M,\text{mne}}_{ss}, \quad \text{where} \quad \pi^{M,\text{mne}}_{ss} = \pi^{\text{MNE}}_{s,\text{U.S.}} \times \pi^{\text{GDP}}_{s,\text{U.S.}}.$$

Finally, we calibrate γ_{state} by minimizing the distance between the observed self share and the model-implied self share of each state. Specifically, we solve the following optimization problem:

$$\min_{\gamma_{\text{state}}} \sum_{s \in \mathcal{S}} \frac{1}{|\mathcal{S}|} \left(\tilde{\pi}^M_{ss}(\mathcal{D}; \gamma_{\text{state}}) - \pi^M_{ss} \right)^2,$$

where $\tilde{\pi}^M_{ss}(\mathcal{D}; \gamma_{\text{state}})$ is the model-implied MP self share for state s, π^M_{ss} is the corresponding observed share, and $|\mathcal{S}|$ is the total number of states. Solving this problem yields the optimal solution γ^*_{state}. Using γ^*_{state} together with the dataset $\mathcal{D}$, all other parameters and MP shares can be recovered by solving the system of equations (1)–(6).

C MP / Trade to GDP Ratio

In this section we report MP /trade to GDP ration between the major economies in the world, supplementing the result in Section 4.

Table C.1 reports the ratio of import to GDP between the major economies in the world in 2005 and 2016, which is the baseline for Table 1.

In 2005, the export of the United States to the EU were almost identical to those of China, while its exports to the Rest of the World (ROW) significantly exceeded China's. However, this pattern

[11] (Kamal et al., 2022) contains data on Share of Establishments by State and Multinational Status, for the year 2017.

had reversed by 2016. The EU's total imports-to-GDP ratio remained the highest in both years, exceeding 30%.

(a) 2005

	USA	CHN	EU	ROW
USA		2.11	1.64	3.21
CHN	1.99		1.64	1.49
EU	2.44	3.17	8.70	7.38
ROW	8.86	23.20	13.40	4.15
Total imports-to-GDP	13.29	28.49	34.07	20.38

(b) 2016

	USA	CHN	EU	ROW
USA		1.18	1.95	3.19
CHN	2.46		2.79	3.92
EU	2.22	1.82	9.30	6.96
ROW	6.96	10.87	14.56	3.83
Total imports-to-GDP	11.64	13.87	37.90	21.74

Table C.1: Import / GDP Ratios (%)

Notes: This table reports the ratio of import to GDP from the source economy (rows) to destination (columns) in different years.

Table C.2 reports the ratio of inward MP to GDP between the major economies in the world in 2005 and 2016, which is the baseline for Table 2. The inward MP from the United States and the EU to all economies are broadly comparable, while China's is substantially smaller — a pattern that differs from the trade data shown in Table C.1. The EU has the highest inward MP-to-GDP ratio in both years, at approximately 40%.

(a) 2005

	USA	CHN	EU	ROW
USA		3.01	9.96	5.69
CHN	0.01		0.02	0.09
EU	7.31	2.57	10.82	5.55
ROW	4.84	7.92	5.09	1.92
Total inward MP-to-GDP	12.16	13.51	36.72	15.17

(b) 2016

	USA	CHN	EU	ROW
USA		2.50	12.35	5.51
CHN	0.17		0.56	0.88
EU	8.13	2.60	12.78	5.11
ROW	6.80	9.39	7.28	2.33
Total inward MP-to-GDP	15.10	14.49	45.75	16.17

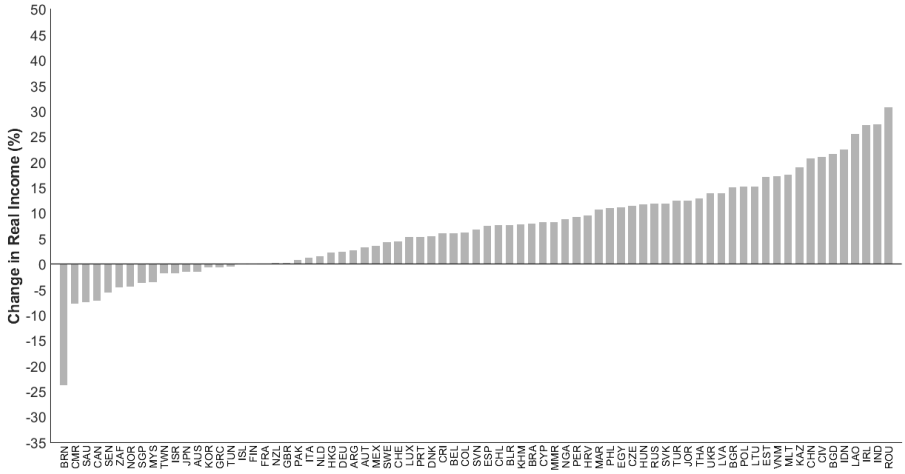
Table C.2: Inward MP / GDP Ratios (%)

Notes: This table reports the ratio of inward MP to GDP from the source economy (rows) to destination (columns) in different years.

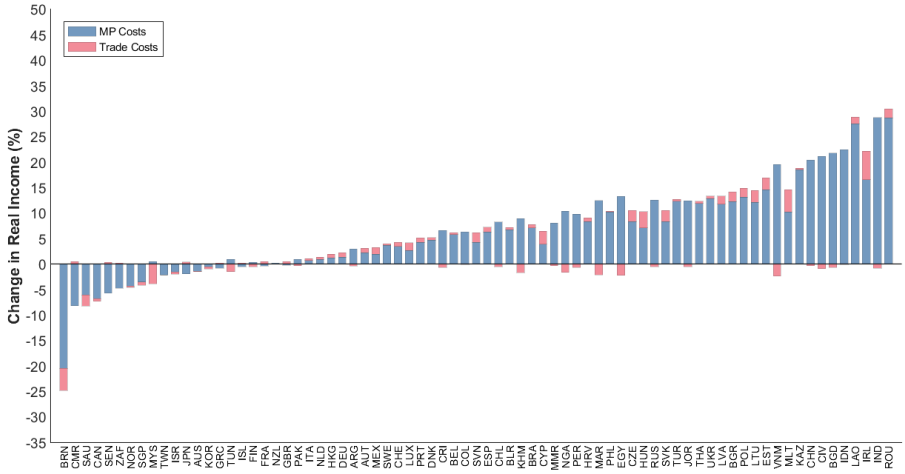
D Welfare Effects of Cost Changes

In this section, we report the cross-country and cross-state distribution of percentage changes in real income resulting from observed changes in MP and trade costs between 2000–2005 and between 2005–2016, supplementing the results presented in Section 5.

Figure D.1 reports the welfare impact of the observed cost changes over 2005–2016 across countries, and Figure D.2 reports the corresponding impact across states. Figures D.3– D.4 respectively present the counterpart for the period 2000–2005. These figures parallel those presented in Figures 4–5. MP cost changes also generate larger welfare effects than trade cost changes during all of these periods, while both two periods are an order of magnitude larger than the counterpart of period 2016-2019. Changes in trade costs exert a larger effect than changes in MP costs for US states throughout all periods.



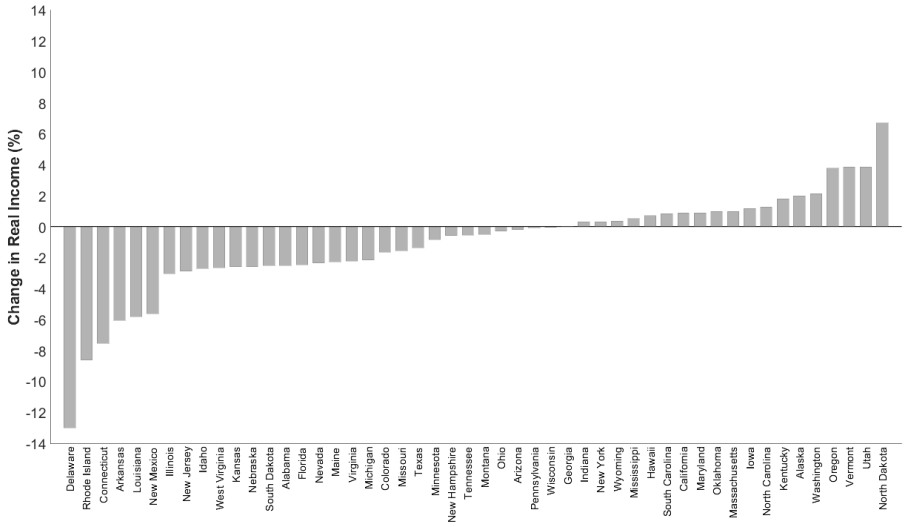
(a) Combined Effects of Changes in MP and Trade Costs



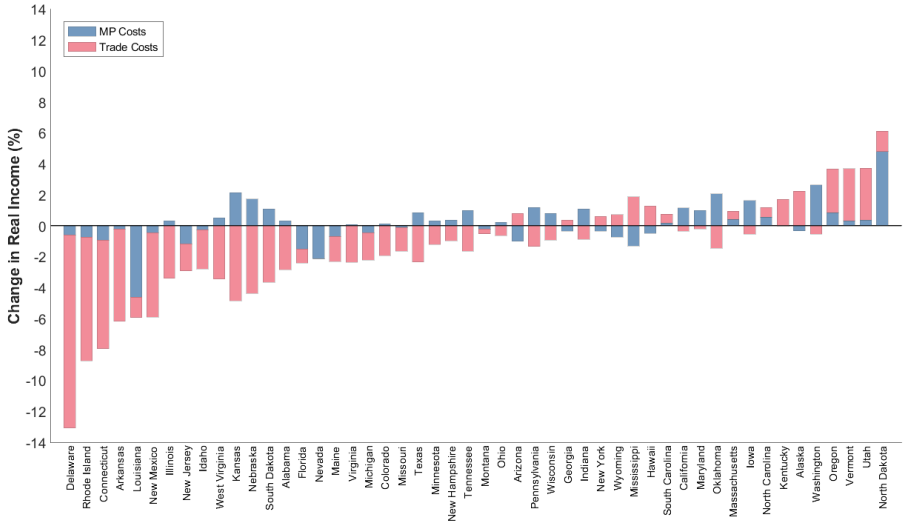
(b) Separate Effects of Changes in MP and Trade Costs

Figure D.1: Gains from Trade and MP Costs Changes, 2005-2016

Notes: Panel (a) shows the combined welfare impact of trade and MP cost changes over 2005-2016 for each economy. Each bar reports the percentage change in real income between the model factual(where costs evolve according to observed data) and the counterfactual scenario (where trade and MP costs are fixed at 2005 levels). In both scenarios, knowledge stocks adjust endogenously. Changes are calculated as: (model factual − counterfactual) / model factual × 100%. Panel (b) shows welfare effects to trade costs and MP costs separately, where each attribution captures the full general equilibrium effect including induced knowledge accumulation. These two bars sum up nearly to the total welfare change shown in panel (a), with residual differences due to nonlinearities.



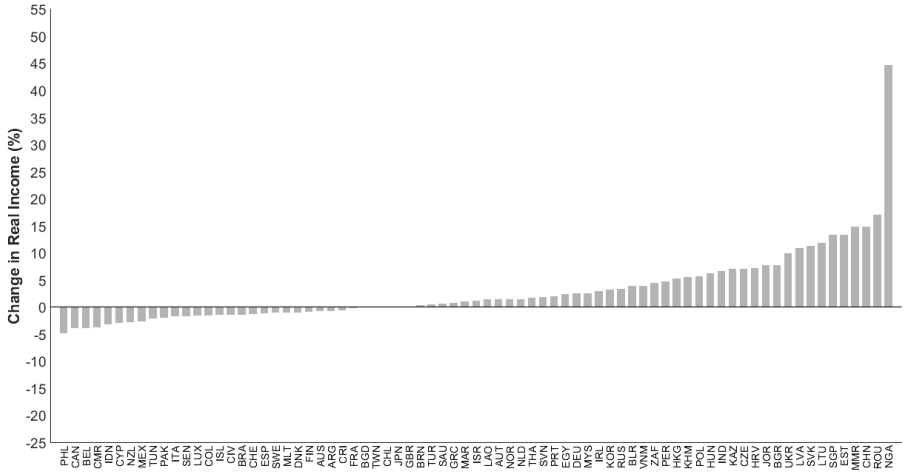
(a) Combined Effects of Changes in MP and Trade Costs



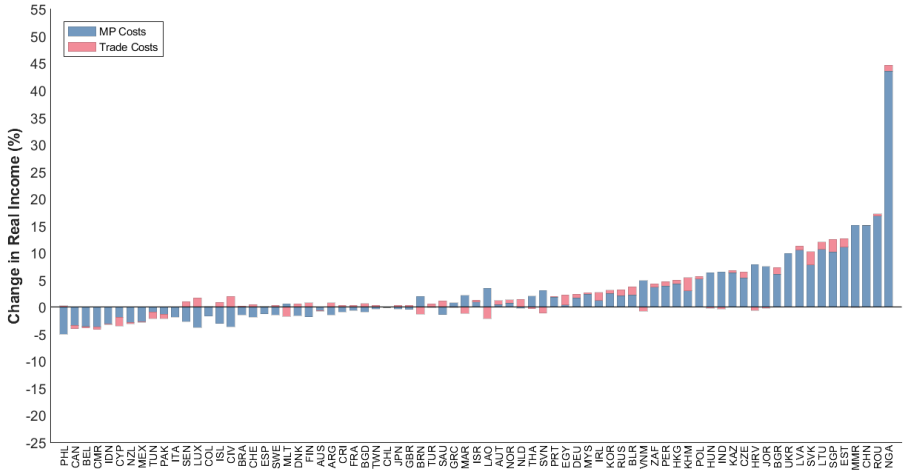
(b) Contributions of Trade and MP Cost Changes

Figure D.2: Gains from Trade and MP Costs Changes, 2005-2016

Notes: Panel (a) shows the combined welfare impact of trade and MP cost changes over 2005-2016 for each state in the US. Each bar reports the percentage change in real income between the model factual (where costs evolve according to observed data) and the counterfactual scenario (where trade and MP costs are fixed at 2005 levels). In both scenarios, knowledge stocks adjust endogenously. Changes are calculated as: (model factual − counterfactual) / model factual × 100%. Panel (b) shows welfare effects to trade costs and MP costs separately, where each attribution captures the full general equilibrium effect including induced knowledge accumulation. These two bars sum up nearly to the total welfare change shown in panel (a), with residual differences due to nonlinearities.



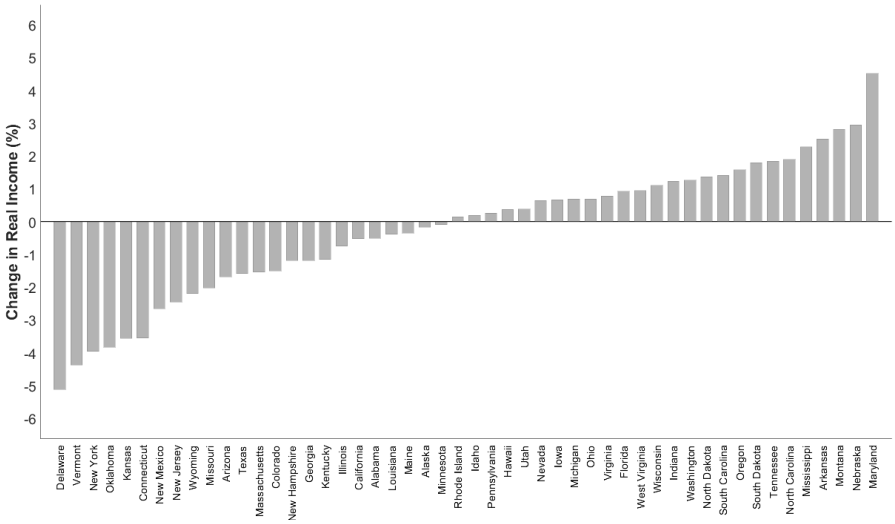
(a) Combined Effects of Changes in MP and Trade Costs



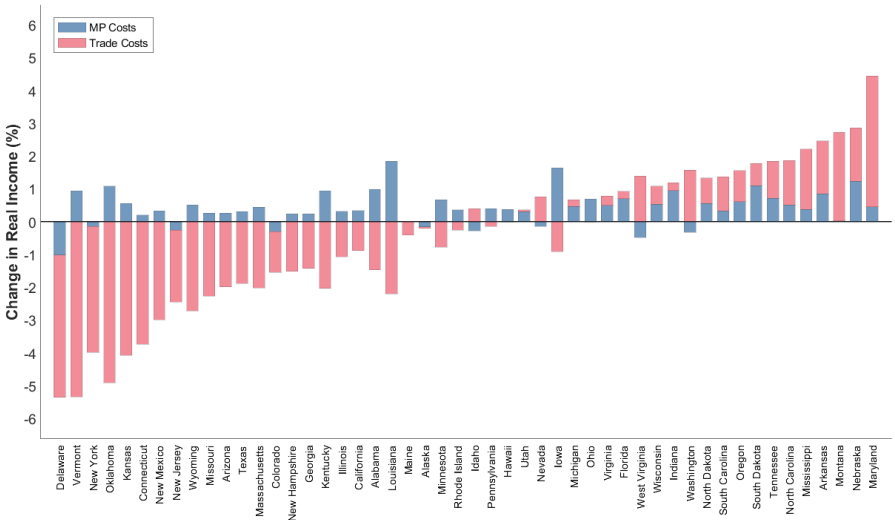
(b) Separate Effects of Changes in MP and Trade Costs

Figure D.3: Gains from Trade and MP Costs Changes, 2000-2005

Notes: Panel (a) shows the combined welfare impact of trade and MP cost changes over 2000-2005 for each economy. Each bar reports the percentage change in real income between the model factual(where costs evolve according to observed data) and the counterfactual scenario (where trade and MP costs are fixed at 2000 levels). In both scenarios, knowledge stocks adjust endogenously. Changes are calculated as: (model factual − counterfactual) / model factual × 100%. Panel (b) shows welfare effects to trade costs and MP costs separately, where each attribution captures the full general equilibrium effect including induced knowledge accumulation. These two bars sum up nearly to the total welfare change shown in panel (a), with residual differences due to nonlinearities.



(a) Combined Effects of Changes in MP and Trade Costs



(b) Contributions of Trade and MP Cost Changes

Figure D.4: Gains from Trade and MP Costs Changes, 2000-2005

Notes: Panel (a) shows the combined welfare impact of trade and MP cost changes over 2000-2005 for each state in the US. Each bar reports the percentage change in real income between the model factual (where costs evolve according to observed data) and the counterfactual scenario (where trade and MP costs are fixed at 2000 levels). In both scenarios, knowledge stocks adjust endogenously. Changes are calculated as: (model factual $-$ counterfactual) / model factual $\times$ 100%. Panel (b) shows welfare effects to trade costs and MP costs separately, where each attribution captures the full general equilibrium effect including induced knowledge accumulation. These two bars sum up nearly to the total welfare change shown in panel (a), with residual differences due to nonlinearities.

E Full List of MP and Trade Integrating Groups

In this section we list by integrating groups for all countries in tables E.1 to E.3.

Economy	MP	Trade	Economy	MP	Trade
BRA	CHN	CHN	IRL	USA	Non
CHN	CHN	CHN	LTU	USA	Non
HKG	CHN	CHN	LVA	USA	Non
JPN	CHN	CHN	POL	USA	Non
MEX	CHN	CHN	ROU	USA	Non
SAU	CHN	CHN	SVK	USA	Non
AUS	CHN	USA	TUR	USA	Non
CAN	CHN	USA	BGD	Non	CHN
CIV	CHN	USA	EGY	Non	CHN
ESP	CHN	USA	JOR	Non	CHN
GBR	CHN	USA	MYS	Non	CHN
ITA	CHN	USA	NGA	Non	CHN
MMR	CHN	USA	PER	Non	CHN
NZL	CHN	USA	SGP	Non	CHN
PAK	CHN	USA	TWN	Non	CHN
ARG	CHN	Non	DNK	Non	USA
BEL	CHN	Non	FIN	Non	USA
BLR	CHN	Non	FRA	Non	USA
CHL	CHN	Non	HUN	Non	USA
CMR	CHN	Non	ISL	Non	USA
COL	CHN	Non	LUX	Non	USA
CRI	CHN	Non	NOR	Non	USA
DEU	CHN	Non	SEN	Non	USA
GRC	CHN	Non	SWE	Non	USA
IDN	CHN	Non	AUT	Non	Non
KAZ	CHN	Non	BRN	Non	Non
KHM	CHN	Non	CHE	Non	Non
LAO	CHN	Non	CYP	Non	Non
MAR	CHN	Non	IND	Non	Non
PHL	CHN	Non	ISR	Non	Non
TUN	CHN	Non	KOR	Non	Non
UKR	CHN	Non	MLT	Non	Non
ZAF	CHN	Non	NLD	Non	Non
USA	USA	USA	PRT	Non	Non
BGR	USA	Non	RUS	Non	Non
CZE	USA	Non	SVN	Non	Non
EST	USA	Non	THA	Non	Non
HRV	USA	Non	VNM	Non	Non

Table E.1: Integrating Groups, 2000-2005

Economy	MP	Trade	Economy	MP	Trade
AUT	CHN	CHN	GBR	Non	CHN
CAN	CHN	CHN	ITA	Non	CHN
CHN	CHN	CHN	MEX	Non	CHN
CIV	CHN	CHN	POL	Non	CHN
FIN	CHN	CHN	TUR	Non	CHN
NOR	CHN	CHN	BGD	Non	USA
SWE	CHN	CHN	BRA	Non	USA
BLR	CHN	USA	CHL	Non	USA
BRN	CHN	USA	COL	Non	USA
KAZ	CHN	USA	CRI	Non	USA
KHM	CHN	USA	EGY	Non	USA
LAO	CHN	USA	IND	Non	USA
TUN	CHN	USA	MAR	Non	USA
UKR	CHN	USA	PER	Non	USA
CHE	CHN	Non	PHL	Non	USA
GRC	CHN	Non	RUS	Non	USA
IRL	CHN	Non	SAU	Non	USA
MMR	CHN	Non	SGP	Non	USA
MYS	CHN	Non	THA	Non	USA
PAK	CHN	Non	ARG	Non	Non
TWN	CHN	Non	AUS	Non	Non
VNM	CHN	Non	BEL	Non	Non
ZAF	CHN	Non	CMR	Non	Non
CYP	USA	USA	HKG	Non	Non
KOR	USA	USA	IDN	Non	Non
MLT	USA	USA	ISR	Non	Non
USA	USA	USA	JOR	Non	Non
ISL	USA	CHN	JPN	Non	Non
BGR	USA	Non	LTU	Non	Non
EST	USA	Non	LUX	Non	Non
HRV	USA	Non	LVA	Non	Non
HUN	USA	Non	NGA	Non	Non
SVK	USA	Non	NLD	Non	Non
CZE	Non	CHN	NZL	Non	Non
DEU	Non	CHN	PRT	Non	Non
DNK	Non	CHN	ROU	Non	Non
ESP	Non	CHN	SEN	Non	Non
FRA	Non	CHN	SVN	Non	Non

Table E.2: Integrating Groups, 2005-2016

Economy	MP	Trade	Economy	MP	Trade
AUS	CHN	CHN	CZE	USA	Non
BEL	CHN	CHN	HRV	USA	Non
CHN	CHN	CHN	LTU	USA	Non
NLD	CHN	CHN	LVA	USA	Non
NZL	CHN	CHN	MLT	USA	Non
SVN	CHN	CHN	POL	USA	Non
ZAF	CHN	CHN	TUR	USA	Non
BGD	CHN	USA	CAN	Non	CHN
CIV	CHN	USA	GBR	Non	CHN
IDN	CHN	USA	MAR	Non	CHN
ITA	CHN	USA	PHL	Non	CHN
MMR	CHN	USA	SEN	Non	CHN
TWN	CHN	USA	CYP	Non	USA
ARG	CHN	Non	DNK	Non	USA
AUT	CHN	Non	FIN	Non	USA
BLR	CHN	Non	IND	Non	USA
CMR	CHN	Non	KOR	Non	USA
CRI	CHN	Non	NGA	Non	USA
DEU	CHN	Non	PER	Non	USA
EGY	CHN	Non	RUS	Non	USA
ESP	CHN	Non	SVK	Non	USA
EST	CHN	Non	BRA	Non	Non
FRA	CHN	Non	BRN	Non	Non
HKG	CHN	Non	CHE	Non	Non
HUN	CHN	Non	COL	Non	Non
JOR	CHN	Non	GRC	Non	Non
KHM	CHN	Non	IRL	Non	Non
LAO	CHN	Non	JPN	Non	Non
MEX	CHN	Non	KAZ	Non	Non
NOR	CHN	Non	LUX	Non	Non
PAK	CHN	Non	MYS	Non	Non
PRT	CHN	Non	SAU	Non	Non
TUN	CHN	Non	SGP	Non	Non
ISL	USA	CHN	SWE	Non	Non
CHL	USA	USA	THA	Non	Non
ROU	USA	USA	UKR	Non	Non
USA	USA	USA	VNM	Non	Non
BGR	USA	Non			

Table E.3: Integrating Groups, 2016-2019

F Counterfactual Results

In this secion we report static and welfare gains from integration group realignment, supplementing the results presented in Section 6.3.

F.1 Split Alignments Between US and China

(a) China MP-Integrating and US Trade-Integrating

	Both Join USA Group		Both Join CHN Group	
Economy	Welfare (%)	Static Gain (%)	Welfare (%)	Static Gain (%)
Bangladesh	0.21	0.01	-0.01	-0.18
Côte d'Ivoire	1.18	0.04	-0.09	-0.15
Indonesia	-1.12	-0.01	-0.13	-0.02
Italy	0.05	-0.08	-0.28	0.03
Myanmar	-2.17	0.01	-0.76	-0.19
Chinese Taipei	-1.18	-0.06	-0.05	-0.08

(b) US MP-Integrating and China Trade-Integrating

	Both Join USA Group		Both Join CHN Group	
Economy	Welfare (%)	Static Gain (%)	Welfare (%)	Static Gain (%)
Iceland	-0.12	-0.03	-1.29	0.12

Table F.1: Gains from Consistent Integration: Split Alignment Economies

Notes: This table presents welfare effects for economies with split integration patterns between the US and China. Panel (a) lists economies that are China MP-integrating but US trade-integrating. Panel (b) lists the economy that is US MP-integrating but China trade-integrating. The first two columns show the effects of both trade and MP joining the US-integrating group. The next two columns show the effects of both joining the China-integrating group. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

Table F.1 reports welfare effects for economies with opposing integration groups in trade and MP. Panel (a) lists economies that are China MP-integrating but US trade-integrating; Panel (b) lists the reverse. Static gains from consolidation are predominantly negative, and welfare effects are typically smaller than static effects, indicating additional negative growth effects.

F.2 China MP-Integrating, Trade Non-Integrating

Table F.2 reports welfare effects for economies that are China MP-integrating but trade non-integrating. The counterfactual assigns them to the China trade-integrating group. Of 20 economies, 13 experience negative static gains. Among those with positive static gains, several see overall welfare turn negative once growth effects are included.

Economy	Welfare (%)	Static Gain (%)
Argentina	-0.02	-0.05
Austria	0.02	0.03
Belarus	-0.15	-0.08
Cameroon	-0.44	-0.42
Costa Rica	0.02	-0.01
Germany	-0.79	0.00
Egypt	-0.06	-0.14
Spain	-0.26	-0.00
Estonia	-0.22	-0.18
France	0.14	0.02
Hong Kong	-0.09	0.02
Hungary	-0.26	-0.09
Jordan	-0.54	-0.97
Cambodia	-0.10	-0.21
Lao	-0.44	-0.21
Mexico	0.00	-0.01
Norway	0.35	0.01
Pakistan	-0.30	-0.07
Portugal	0.10	0.00
Tunisia	-0.03	0.02

Table F.2: Gains from Joining China Trade-Integrating Group

Notes: This table shows welfare effects for economies that are China MP-integrating but trade non-integrating. The counterfactual examines the effects of these economies joining the China trade-integrating group, thereby achieving consistent China integration in both dimensions. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

F.3 China Trade-Integrating, MP Non-Integrating

Table F.3 reports welfare effects for economies that are China trade-integrating but MP non-integrating. The counterfactual assigns them to the China MP-integrating group. All economies experience negative welfare effects, consistent with trade integration with China not implying complementarities in production networks.

F.4 US MP-Integrating, Trade Non-Integrating

Table F.4 reports welfare effects for economies that are US MP-integrating but trade non-integrating. The counterfactual assigns them to the US trade-integrating group. A majority experience negative welfare effects, reinforcing the finding that forced consolidation typically reduces welfare.

Economy	Welfare (%)	Static Gain (%)
Canada	-0.18	-0.00
United Kingdom	-0.96	0.00
Morocco	-0.29	-0.03
Philippines	-0.37	-0.08
Senegal	-0.73	0.01

Table F.3: Gains from Joining China MP-Integrating Group

Notes: This table shows welfare effects for economies that are China trade-integrating but MP non-integrating. The counterfactual examines the effects of these economies joining the China MP-integrating group, thereby achieving consistent China integration in both dimensions. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

F.5 US Trade-Integrating, MP Non-Integrating

Table F.5 reports welfare effects for economies that are US trade-integrating but MP non-integrating. The counterfactual assigns them to the US MP-integrating group. Results are mixed: most economies achieve positive welfare gains, while others, particularly South Korea and India, face losses.

F.6 Tariff-Only Counterfactual

The following tables report results from a variant of the consistent integration counterfactual in which only the tariff component of trade costs is adjusted, leaving non-tariff iceberg costs and MP costs at their actual values.

Table F.6 reports tariff-only results for economies with split alignments between China and the US. For each economy, trade integration group assignment is changed to be consistent with its MP integration group by adjusting tariffs only. Table F.7 reports tariff-only results for economies that are China MP-integrating but trade non-integrating. Table F.8 reports tariff-only results for economies that are US MP-integrating but trade non-integrating.

Economy	Welfare (%)	Static Gain (%)
Bulgaria	0.02	0.02
Czech Republic	-0.25	-0.11
Croatia	-0.31	-0.07
Israel	-0.19	-0.23
Lithuania	-0.07	-0.08
Latvia	0.12	0.04
Malta	0.47	-0.07
Poland	-0.13	-0.01
Turkey	0.38	0.09

Table F.4: Gains from Joining US Trade-Integrating Group

Notes: This table presents welfare effects for economies that are US MP-integrating but trade non-integrating. The counterfactual examines the effects of these economies joining the US trade-integrating group, thereby achieving consistent US integration in both dimensions. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

Economy	Welfare (%)	Static Gain (%)
Cyprus	1.89	-2.05
Denmark	0.52	-0.14
Finland	0.49	0.01
India	-0.20	0.01
South Korea	-1.43	-0.03
Nigeria	0.05	0.00
Peru	0.36	0.06
Russia	1.23	0.02
Slovakia	0.38	-0.21

Table F.5: Gains from Joining US MP-Integrating Group

Notes: This table presents welfare effects for economies that are US trade-integrating but MP non-integrating. The counterfactual examines the effects of these economies joining the US MP-integrating group, thereby achieving consistent US integration in both dimensions. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

Economy	Welfare (%)	Static Gain (%)
Bangladesh	0.05	-0.02
Côte d'Ivoire	0.04	-0.01
Indonesia	-0.05	0.01
Italy	-0.06	0.00
Myanmar	0.14	0.02
Chinese Taipei	0.01	0.00
Iceland	-0.12	0.01

Table F.6: Tariff-Only: Gains from Consistent Integration

Notes: This table presents welfare effects for economies with split integration patterns–China MP-integrating but US trade-integrating (above the mid-rule) and US MP-integrating but China trade-integrating (below). For each economy, we realign trade integration to be consistent with its MP integration group by changing tariffs only. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

Economy	Welfare (%)	Static Gain (%)
Argentina	0.11	0.15
Austria	0.03	0.02
Belarus	-0.01	0.06
Cameroon	-0.07	-0.06
Costa Rica	-0.02	-0.01
Germany	-0.22	-0.01
Egypt	-0.28	-0.21
Spain	-0.01	-0.01
Estonia	-0.00	-0.01
France	-0.05	-0.01
Hong Kong	-0.13	0.01
Hungary	-0.05	-0.04
Jordan	-0.02	-0.01
Cambodia	-0.07	-0.09
Lao	0.04	0.01
Mexico	-0.02	-0.01
Norway	0.00	-0.01
Pakistan	0.22	0.21
Portugal	0.04	-0.00
Tunisia	-0.01	-0.07

Table F.7: Tariff-Only: Gains from Joining China Trade-Integrating Group

Notes: This table shows tariff-only welfare effects for economies that are China MP-integrating but trade non-integrating. The counterfactual assigns them to the China trade-integrating group by adjusting tariffs only. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

Economy	Welfare (%)	Static Gain (%)
Bulgaria	0.01	0.03
Czechia	-0.03	0.01
Croatia	0.02	0.00
Israel	-0.01	0.00
Lithuania	-0.01	0.02
Latvia	-0.02	0.01
Malta	-0.02	-0.02
Poland	-0.04	0.00
Turkey	0.01	-0.01

Table F.8: Tariff-Only: Gains from Joining US Trade-Integrating Group

Notes: This table reports tariff-only welfare effects for economies that are US MP-integrating but trade non-integrating. The counterfactual assigns them to the US trade-integrating group by adjusting tariffs only. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

G Supplemental Exercises

In this section, we perform counterfactual exercises to examine whether tariff or non-tariff components dominate during the realignment of trade integration groups, supplementing the exercises in Section 6.3.3.

G.1 Change Non-Tariff Trade Costs Only

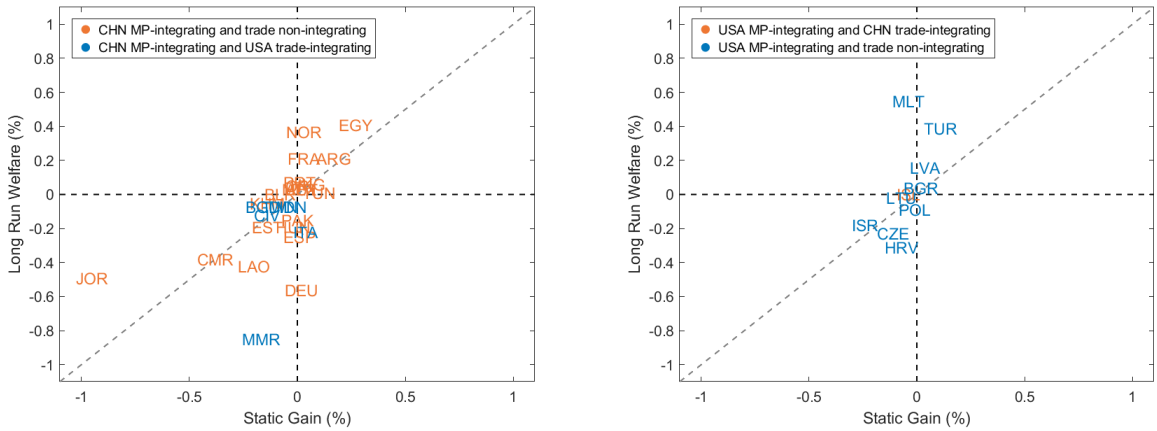


Figure G.1: Welfare Effects of Integration Group Realignment

Notes: This figure displays static and welfare gains from integration group realignment. The left panel presents results for economies switching to full China-integrating status. For economies initially China MP-integrating but not China trade-integrating, this entails switching to the China trade-integrating group. For economies initially China trade-integrating but not China MP-integrating, this entails switching to the China MP-integrating group. The right panel presents analogous results for economies switching to full US-integrating status. We only change the non-tariff component of trade costs when switching trade integration groups. Welfare represents the annual consumption equivalent variation. Static gain measures the change in real consumption level in 2019 relative to the baseline.

Figure G.1 shows the result compactly for counterfactuals where countries switch trade integation. The graph is similar to Figure 11, complementing the results in section F.6 that changes in the non-tariff component of trade costs seem to be more important during the period from 2016 to 2019.